

SAMA7G5 Series System-in-Package (SiP) MPUs with Embedded DDR3L SDRAM

SAMA7G5 Series SiP Data Sheet



[Product Page Links](#)

Scope

This document is an overview of the main features of the SAMA7G5 Series SiP microprocessor. The sole reference documents for product information on the SAMA7G5 Series devices and the DDR3L SDRAM memories are listed in [Reference Documents](#).

Introduction

The SAMA7G5 Series SiP integrates the Arm® Cortex®-A7 processor-based SAMA7G54 MPU with a DDR3L SDRAM. By combining the SAMA7G54 with a DDR3L SDRAM in a single package, PCB routing complexity, area and number of layers are reduced in the majority of cases. This makes board design easier and more robust by facilitating design for EMI, ESD and signal integrity.

The devices are available in junction temperature ranges up to 125°C and are AEC-Q100 grade 2 qualified.

The following DDR3L SDRAM memory densities are available:

- 1-Gbit DDR3L SDRAM
- 2-Gbit DDR3L SDRAM
- 4-Gbit DDR3L SDRAM

Reference Documents

Type	Name	Literature No./Link
Data sheet	SAMA7G5 Series	DS60001765
Errata sheet	SAMA7G5 Series Silicon Errata and Data Sheet Clarification	DS80001016
Application note	SAMA7G54 Hardware Design Considerations	DS00004598
Data sheet	1-Gbit 8M x 8 BANKS x 16 BIT DDR3L SDRAM	W631GU6NB12 (TFBGA427), W631GU6RB12 (TFBGA442)
Data sheet	2-Gbit 16M x 8 BANKS x 16 BIT DDR3L SDRAM	W632GU6NB12 (TFBGA427), W632GU6RB12 (TFBGA442)
Data sheet	4-Gbit 32M x 8 BANKS x 16 BIT DDR3L SDRAM	W634GU6RB12 (TFBGA427, TFBGA442)

1. Features

- Arm Cortex-A7 Core
 - Arm TrustZone®
 - Arm Neon™ multimedia architecture
 - Floating Point Unit
 - Embedded Trace module with instruction trace stream, including 16 Kbytes of Arm CoreSight™ Embedded Trace buffer
 - 32 Kbytes of L1 data cache, 32 Kbytes of L1 instruction cache
 - 256 Kbytes of L2 cache
 - Up to 1 GHz operating frequency
 - Voltage and frequency scaling support
 - 64-bit generic timers
- Internal Memory Architecture
 - 128 Kbytes of internal SRAM
 - 80 Kbytes of maskable ROM, embedding a secure bootloader (boot on QSPI NOR, SD, eMMC)
 - 96 Kbytes ROM for NAND Flash ECC tables
 - 40 Kbytes ROM for crypto-libraries (RSA, ECC, etc.)
 - 11 Kbytes of internal OTP
 - 16-bit high-bandwidth, DDR3L SDRAM up to 533 MHz, up to 4 Gbits
- External Memory Support
 - 16-bit static memory controller, FPGA support with synchronous clock
 - 8-bit SLC and MLC NAND controller with up to 32-bit error correcting code
 - One 8-bit high-speed memory card host eMMC 5.1 (HS400), SD3.0 SDR104 mode support
 - Two 4-bit high-speed memory card hosts eMMC 4.51 (HS200), SD3.01 SDR104 mode support
 - One octal Serial Peripheral Interface running up to 200 MHz DDR
 - One quad Serial Peripheral Interface
- System
 - Power-on reset cells, reset controller, shutdown controller, Watchdog and secure Watchdog Timers running on internal slow RC oscillator (32 kHz typical) and real-time clock running on a slow crystal oscillator (32.768 kHz)
 - Two internal trimmed RC oscillators with typical values: 32 kHz and 12 MHz
 - Two crystal oscillators: 32.768 kHz and 20 to 50 MHz
 - Eight PLLs for core, system bus and peripherals, serial interfaces, DDR I/Os, pixel clock, audio, USB, MIPI CSI-2 and Ethernet
 - Two 32-channel DMAs with per-channel security configuration
 - One 8-channel DMA dedicated to memory-to-memory transactions
 - Eight programmable clock output signals
- Power Considerations
 - Different power domains and modes to reduce power consumption
 - Low-power consumption in Backup mode with 5 Kbytes of secure backup SRAM and DDR-SDRAM in Self-Refresh mode

- Low-power with SRAM and register retention, wake-up from various events (USB, CAN, Ethernet WOL, FLEXCOMs), internal events (RTC, timer) and I/O activity
- Embedded LDOs for MIPI CSI-2, analog and PLLs, to enable low-cost power management solutions
- Optimum connection to Microchip MCP16501/2 PMICs to enter and exit various power modes of the application
- Multimedia Peripherals
 - Audio
 - Two synchronous serial controllers, each with 16 channels of up to 32-bit TDM data
 - One inter-IC sound multi-channel controller with TDM256 support
 - Up to two 4-channel pulse density microphone controllers; support for eight microphones in parallel
 - One Sony/Philips digital interface transmitter and receiver
 - Audio sample rate converter including four stereo channels
 - Image
 - Image sensor controller, ITU-R BT. 601/656 supporting up to eight megapixels for still images and 60 fps in 720p mode, 8 bits, raw Bayer, YCbCr, monochrome, camera ISP
 - 2-lane MIPI CSI-2 (D-PHY) and 12-bit RGB interface support
- Peripherals
 - Two high-speed USB devices and three high-speed USB hosts sharing three on-chip transceivers
 - One 10/100/1000 Gigabit Ethernet MAC supporting RGMII, MII and RMII (GMAC0) and one 10/100 Ethernet MAC supporting MII and RMII (GMAC1) compliant with:
 - IEEE802.3az Energy-Efficient Ethernet
 - IEEE802.1AS Timestamping for Ethernet AVB support
 - IEEE802.1Qav Credit-based traffic shaping hardware support
 - IEEE1588 Precision Time Protocol
 - IEEE1588 Timestamp Unit (TSU) with TSU timer comparison signal triggering a timer counter and available on a PIO line
 - Six flexible data rate CAN-FD controllers with SRAM-based mailboxes with time- and event-triggered transmission
 - Twelve FLEXCOMs (USART, SPI and TWIHS)
 - Six 64-bit timers
 - Two three-channel 32-bit timer counters with PWM generation
 - One four-channel 16-bit PWM controller
 - One 16-channel 12-bit analog-to-digital converter, up to 1 Msps
- Safety
 - Temperature and core voltage monitoring
 - Zero-power power-on reset cells
 - Main crystal monitor and clock failure detector with failsafe switchover to main RC oscillator
 - 32 kHz crystal monitor and clock failure detector with failsafe switchover to internal 32 kHz RC oscillator
 - Integrity check monitor based on SHA256
 - Safety critical modules (WDT, RSTC, SHDWC, etc.) running on always-on slow RC oscillator

- Register write protection
- Security
 - TrustZone support
 - One Secure TrustZone Watchdog Timer running on RC oscillator, providing protection against TrustZone starvation
 - Temperature, voltage and frequency monitoring
 - Secure backup SRAM
 - 5 Kbytes scrambled with non-imprinting support powered with VBAT or VDDIN33:
 - 1 Kbyte non-erasable on tamper detection
 - 4 Kbytes erasable on tamper detection
 - Four tamper pins for static or dynamic detection
 - Can be used as regular wake-up lines
 - 256-bit general purpose backup register, erasable on tamper detection
 - Programmable OTP with bits available for user purposes
 - Configurable JTAG/SWD security (full debug, non-secure-only debug, no debug)
 - 128-bit AES on-the-fly encryption/decryption on DDR memory, SMC, QSPI0 and QSPI1, including automatic key load at start-up. Separate keys for secure and non-secure accesses (TZAESB).
 - True random number generator compliant with NIST Special Publication 800-22 Tests Suite and FIPS PUB 140-2 and 140-3
 - Secure RTC
- Cryptography
 - SHA (SHA1, SHA224, SHA256, SHA384, SHA512) compliant with FIPS Publications 180-2
 - AES: 256-, 192-, 128-bit key algorithm, compliant with FIPS PUB 197 specifications
 - TDES: two-key or three-key algorithms, compliant with FIPS PUB 46-3 specifications
 - Public Key Coprocessor (CPKCC) and associated Classical Public Key Cryptography Library (CPKCL) for RSA, DSA, ECC GF(2ⁿ), ECC GF(p)
- Up to 136 I/Os
 - Fully programmable through set/clear registers
 - Multiplexing of eight peripheral functions per I/O line
 - Each I/O line can be assigned to a peripheral or used as a general purpose I/O
 - PIO controller featuring a synchronous output providing up to 32 bits of data output in a single write operation
- Design for Low ElectroMagnetic Interference (EMI)
 - Slew rate controlled I/Os
 - DDR PHY with impedance-calibrated drivers
 - Spread spectrum PLLs
 - Careful BGA power/ground ball assignment to provide optimum decoupling capacitors placement
- Microchip Recommended Power Management Integrated Circuits (PMICs)
 - MCP16502, 6-channel PMIC with I²C control interface; supports dynamic voltage scaling and processor Low-Power modes (ULP2, BSR)
 - MCP16501, 4-channel PMIC optimized for compact PCB layout; supports processor Low-Power mode

- Operating Conditions
 - Junction temperature ranges (T_J):
 - -40°C to $+105^{\circ}\text{C}$ (Industrial)
 - -40°C to $+125^{\circ}\text{C}$ (Extended Industrial and Automotive)
- Qualification
 - AEC-Q100 Grade 2 qualified (-E/XHBVAO devices only) using the following mission profile:
 - 20 khrs of operation with junction temperature usage as $-40^{\circ}\text{C} = 6\%$ of lifetime, $70^{\circ}\text{C} = 25\%$, $85^{\circ}\text{C} = 25\%$, $100^{\circ}\text{C} = 25\%$, $115^{\circ}\text{C} = 14\%$, $125^{\circ}\text{C} = 5\%$
 - Maximum CPU speed = 800 MHz
 - ESD-CDM classification level C2
 - The AEC-Q006 set of tests applies, as only copper wire interconnections are used.
- Packages
 - 427-ball TFBGA 18x21x1.2 mm, 0.8 mm pitch
 - 442-ball TFBGA 16x16x1.2 mm, 0.5 mm pitch optimized for standard class PCB layout

2. Ordering Information

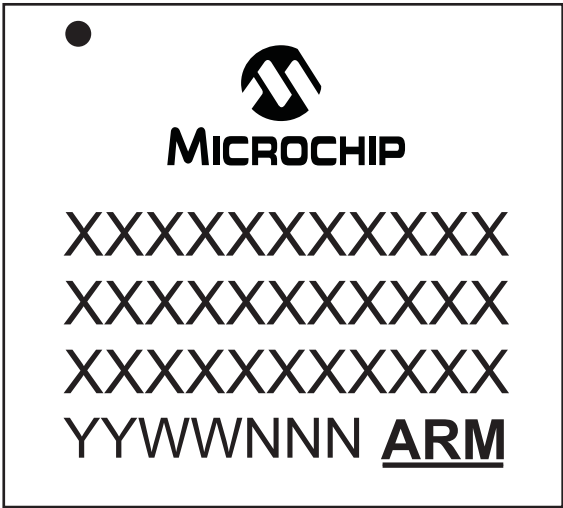
Ordering Code ⁽¹⁾⁽²⁾	Memory Type	Memory Size	Tier	Max CPU Speed	Package	Junction Temperature Range ⁽³⁾		
SAMA7G54D1G(T)-I/XHB(-SLx)	DDR3L SDRAM	1 Gbit	Industrial	1 GHz	TFBGA442	-40°C to +105°C		
SAMA7G54D1G(T)-I/4UB(-SLx)					TFBGA427			
SAMA7G54D2G(T)-I/XHB(-SLx)		2 Gbits			TFBGA442			
SAMA7G54D2G(T)-I/4UB(-SLx)					TFBGA427			
SAMA7G54D4G(T)-I/XHB(-SLx)		4 Gbits			TFBGA442			
SAMA7G54D4G(T)-I/4UB(-SLx)					TFBGA427			
SAMA7G54D1G(T)-E/XHB(-SLx)		1 Gbit	Extended Industrial	1 GHz ⁽⁴⁾	TFBGA442	-40°C to +125°C		
SAMA7G54D2G(T)-E/XHB(-SLx)		2 Gbits						
SAMA7G54D4G(T)-E/XHB(-SLx)		4 Gbits						
SAMA7G54D1G(T)-E/XHB(-SLx)VAO		1 Gbit	Automotive	800 MHz				
SAMA7G54D2G(T)-E/XHB(-SLx)VAO		2 Gbits						
SAMA7G54D4G(T)-E/XHB(-SLx)VAO		4 Gbits						

Notes:

- For details on ordering codes, see [Product Identification System](#).
- For SL1, SL2 and SL3 device availability, contact a Microchip Sales representative.
- Applies to both the MPU and the DDR3L memory junction temperatures.
- 1 GHz up to T_j = 105°C, 800 MHz for T_j in the [105°C - 125°C] range

3. Marking

Top marking follows the scheme below:

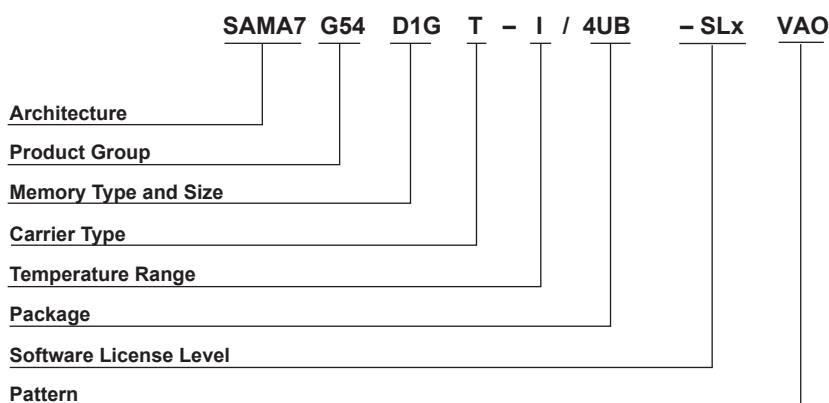


with possible values:

Line	Description	Values
1	Company logo	Microchip logo
2	Company name	Microchip
3	Device name	SAMA7G54-DxG
4	Temperature code/Package	I/4UB  I/XHB  E/XHB 
5	Not used	-
6	Lot traceability, Arm logo	YYWWNNN ARM

4. Product Identification System

To order or obtain information, for example, on pricing or delivery, contact Microchip: <https://www.microchip.com/en-us/about/contact-us>.



Architecture:	SAMA7	= Arm Cortex-A7 CPU
Product Group:	G54	= 427-ball general-purpose microprocessors
Memory Type and Size:	D1G	= 1-Gbit DDR3L SDRAM
	D2G	= 2-Gbit DDR3L SDRAM
	D4G	= 4-Gbit DDR3L SDRAM
Carrier Type ⁽¹⁾ :	Blank	= Standard packaging (tray)
	T	= Tape and Reel
Ambient Temperature Range ⁽²⁾ :	I	= -40°C to +85°C (industrial)
	E	= -40°C to +105°C (extended industrial or automotive (AEC-Q100 grade 2))
Package:	4UB	= 427-ball TFBGA
	XHB	= 442-ball TFBGA
Software License Level ⁽¹⁾ :	Blank	= Standard
	SL1	= Level 1
	SL2	= Level 2
	SL3	= Level 3
Pattern:	Blank	= Standard device
	VAO	= Automotive flow/material without PPAP. Available upon request.
	Vxx	= Automotive flow/material with associated PPAP. Available upon request.
Note: Not all combinations are valid. See Ordering Information for the list of valid ordering codes.		

Notes:

1. The Tape and Reel identifier and the Software License Level identifier only appear in the catalog part number description. These identifiers are used for ordering purposes and are not printed on the device package. Check with your Microchip Sales Office for package availability.
2. Indicative Ambient Temperature Range. The user must not exceed the maximum Junction Temperature (T_j) defined in the [Electrical Characteristics](#) section.

5. DDR3L SDRAM Features

The SAMA7G5 Series SiP is available with 1-Gbit, 2-Gbit or 4-Gbit DDR3L SDRAM memory options. For power consumption, electrical characteristics and timings of these memories, refer to the manufacturers' data sheets listed in [Reference Documents](#).

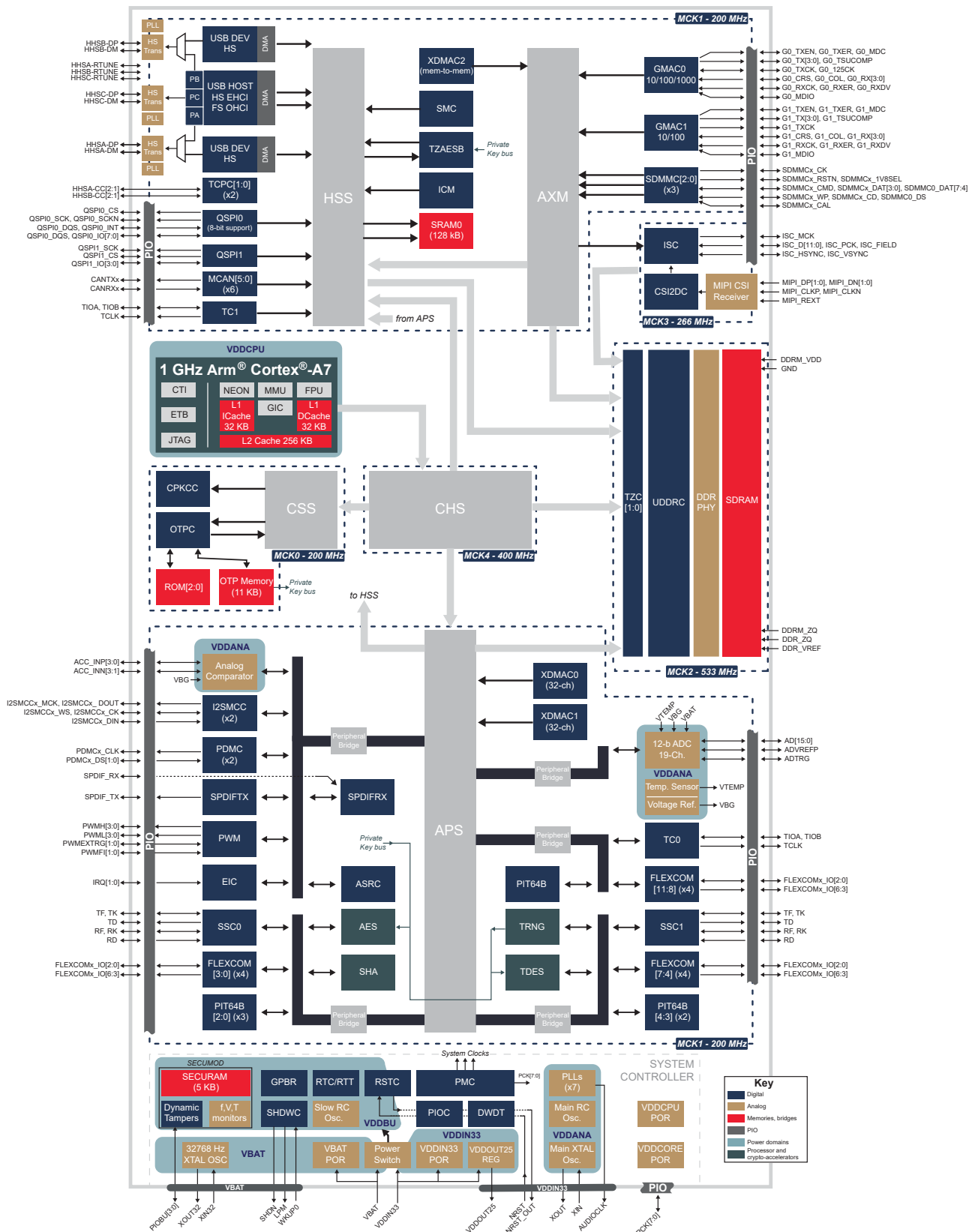
- Power supply: DDR3L DDRM_VDD = 1.283V to 1.45V
- 2-Kbyte page size (x16)
- 8-bank operation controlled by BA0, BA1 and BA2
- Burst lengths (BL): 8 and 4 with Burst Chop (BC)
- Precharge: auto-precharge option for each burst access
- Refresh: auto-refresh, self-refresh
- High-speed data transfer realized by the 8-bit prefetch pipelined architecture
- Double Data Rate architecture: two data transfers per clock cycle
- Bidirectional differential data strobe (DQS and /DQS) transmitted/received with data for capturing data at the receiver
- DQS edge-aligned with data for reads and center-aligned with data for writes
- Differential clock inputs (CK and /CK)
- DLL aligns DQ and DQS transitions with CK transitions
- Commands entered on each positive CK edge; data and data mask referenced to both edges of DQS
- Data mask (DM) for write data
- Posted CAS by programmable additive latency for better command and data bus efficiency
- MultiPurpose Register (MPR) for predefined pattern read out
- ZQ calibration for DQ drive and ODT
- Programmable Partial Array Self-Refresh (PASR)
- Reset pin for power-up sequence and reset function
- Self-Refresh Temperature (SRT) range: normal/extended
- Automatic Self-Refresh (ASR)
- Programmable output driver impedance control

6. Configuration Summary

Feature	SAMA7G54D1G	SAMA7G54D2G	SAMA7G54D4G
Package	TFBGA427 TFBGA442		
CPU	Cortex-A7		
CPU frequency	Up to 1 GHz		
External memory support	NAND Flash, QSPI (NAND, NOR)		
Number of PIOs	136		
SDMMC	3		
DDR datapath	16-bit, 533 MHz (internal)		
Internal SDRAM	1-Gbit DDR3L SDRAM	2-Gbit DDR3L SDRAM	4-Gbit DDR3L SDRAM
GMAC	RGMII/MII/RMII + MII/RMII		
CAN	6		
FLEXCOM (USART/SPI/I2C)	12		
ADC channels	16		
USB device/host	2/2 sharing 2 USB Type-C™ transceivers + 1 host		
ISC	MIPI + Parallel RGB		
I2SMCC channel outputs/inputs	8/8		
SSC	2		
PDMC channels	Up to 8 microphones		
SPDIF	RX + TX		
Audio sample rate converter	1		
QSPI	Octal + Quad		
64-bit timers/32-bit timers	6/2		
PWM	4 differential signals, 2 external triggers, 2 fault inputs		
Cryptography	PKCC, AES, SHA, TRNG, TDES		

7. Block Diagram

Figure 7-1. SAMA7G5 Series SiP Block Diagram



8. Chip Identifier

Table 8-1. Chip ID and Extended ID Definition

Chip Name	CHIPID_CIDR	CHIPID_EXID ⁽¹⁾
SAMA7G54D1G	0x8016211x	0x00000018 0x000000CC (SL1) 0x000000CD (SL2) 0x000000CE (SL3)
SAMA7G54D2G		0x00000020 0x000000D0 (SL1) 0x000000D1 (SL2) 0x000000D2 (SL3)
SAMA7G54D4G		0x00000028 0x000000D4 (SL1) 0x000000D5 (SL2) 0x000000D6 (SL3)

Note:

1. For details on ordering codes, see [Product Identification System](#).

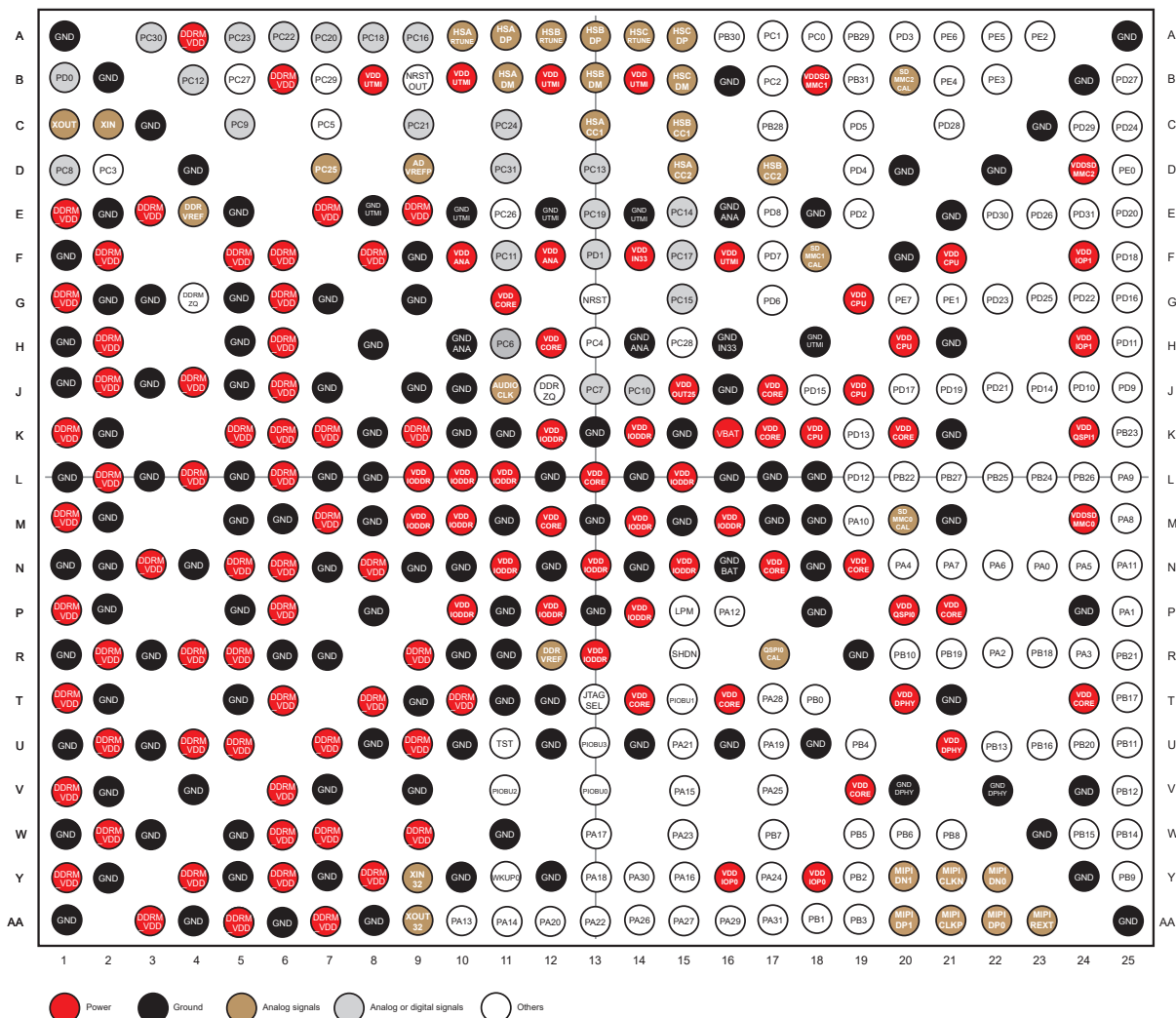
9. Package and Pinout

All SAMA7G5 Series SiP devices using the same package are pin-to-pin compatible.

Package Name	Ball Count	Ball Pitch	Package Size
TFBGA427	427	0.8 mm	18x21x1.2 mm
TFBGA442	442	0.5 mm	16x16x1.2 mm

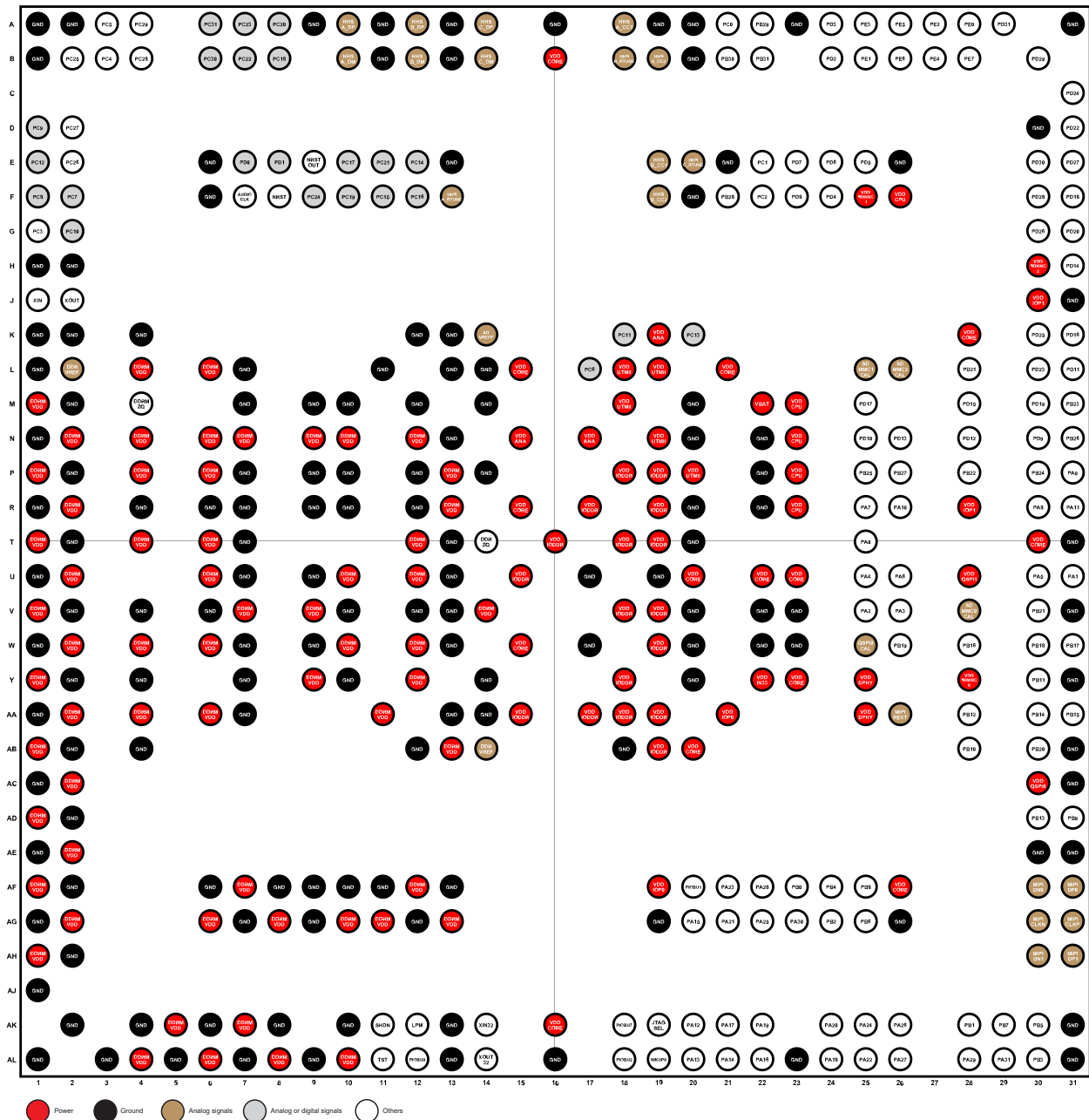
9.1. TFBGA427 Package

Figure 9-1. 427-Ball TFBGA Pinout



9.2. TFBGA442 Package

Figure 9-2. 442-Ball TFBGA Pinout



9.3. Pin Description

The device features several PIO controllers that multiplex the I/O lines of the peripheral set. [Table 9-1](#) defines how the I/O lines are multiplexed on the different PIO controllers. The "Reset State" column shows whether the PIO line resets in I/O mode or in Peripheral mode. If I/O is shown, the PIO line resets with the characteristics (input, output, pull-up or pull-down) indicated in this same column, so that the device is configured in a known state as soon as the reset is released. As a result, PIO_CFGR.FUNC resets to '0'. If a signal name is shown in the "Reset State" column, the PIO line is assigned to this function and PIO_CFGR.FUNC is not set to '0'. That is the case for pins controlling memories, particularly address lines, which require the pin to be driven as soon as the reset is released.

Table 9-1. Pin Description⁽¹⁾

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
N23	T25	VDDSDMMC0	HSIO	PA0	I/O	-	-	A	SDMMC0_CK	O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO0	I/O	1	
								C	CANTX3	O	1	
								E	PWML0	O	3	
P25	U31	VDDSDMMC0	HSIO	PA1	I/O	-	-	A	SDMMC0_CMD	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO1	I/O	1	
								C	CANRX3	I	1	
								D	D14	I/O	1,2	
R22	V25	VDDSDMMC0	GPIO	PA2	I/O	-	-	E	PWMH0	O	3	PIO, I, PU, ST
								A	SDMMC0_RSTN	O	1	
								B	FLEXCOM0_IO2	I/O	1	
								C	PDMC1_CLK	O	1	
R24	V26	VDDSDMMC0	HSIO	PA3	I/O	-	-	D	D15	I/O	1,2	PIO, I, PU, ST
								E	PWMH1	O	3	
								F	FLEXCOM1_IO0	I/O	3	
								A	SDMMC0_DAT0	I/O	1	
N20	U25	VDDSDMMC0	HSIO	PA4	I/O	-	-	B	FLEXCOM0_IO3	I/O	1	PIO, I, PU, ST
								C	PDMC1_DS0	I	1	
								D	NWR1/NBS1	O	1,2	
								E	PWML3	O	3	
N24	U30	VDDSDMMC0	HSIO	PA5	I/O	-	-	F	FLEXCOM1_IO1	I/O	3	PIO, I, PU, ST
								A	SDMMC0_DAT1	I/O	1	
								B	FLEXCOM0_IO4	O	1	
								C	PDMC1_DS1	I	1	
								D	NCS2	O	1,2	PIO, I, PU, ST
								E	PWMH3	O	3	
								F	FLEXCOM2_IO0	I/O	3	
								A	SDMMC0_DAT2	I/O	1	
								B	FLEXCOM1_IO0	I/O	1	PIO, I, PU, ST
								C	CANTX2	O	1	
								D	A23	O	1,2	
								E	PWMEXTRG0	I	3	
								F	FLEXCOM2_IO1	I/O	3	PIO, I, PU, ST

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
N22	U26	VDDSDMMC0	HSIO	PA6	I/O	-	-	A	SDMMC0_DAT3	I/O	1	PIO, I, PU, ST
								B	FLEXCOM1_IO1	I/O	1	
								C	CANRX2	I	1	
								D	A24	O	1,2	
								E	PWMEXTRG1	I	3	
								F	FLEXCOM3_IO0	I/O	3	
N21	R25	VDDSDMMC0	HSIO	PA7	I/O	-	-	A	SDMMC0_DAT4	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO0	I/O	1	
								C	CANTX1	O	1	
								D	NWAIT	I	1,2	
								E	PWMFI0	I	3	
								F	FLEXCOM3_IO1	I/O	3	
M25	R30	VDDSDMMC0	HSIO	PA8	I/O	-	-	A	SDMMC0_DAT5	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO1	I/O	1	
								C	CANRX1	I	1	
								D	NCS0	O	1,2	
								E	PWMFI1	I	3	
								F	FLEXCOM4_IO0	I/O	3	
L25	P31	VDDSDMMC0	HSIO	PA9	I/O	-	-	A	SDMMC0_DAT6	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO2	I/O	1	
								C	CANTX0	O	1	
								D	SMCK	O	1,2	
								E	SPDIF_RX	I	1	
								F	FLEXCOM4_IO1	I/O	3	
M19	R26	VDDSDMMC0	HSIO	PA10	I/O	-	-	A	SDMMC0_DAT7	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO3	I/O	1	
								C	CANRX0	I	1	
								D	NCS1	O	1,2	
								E	SPDIF_TX	O	1	
								F	FLEXCOM5_IO0	I/O	3	
N25	R31	VDDSDMMC0	HSIO	PA11	I/O	-	-	A	SDMMC0_DS	I	1	PIO, I, PU, ST
								B	FLEXCOM2_IO4	O	1	
								D	A0/NBS0	O	1,2	
								E	TIOA0	I/O	1	
								F	FLEXCOM5_IO1	I/O	3	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
P16	AK20	VDDIOP0	GPIO	PA12	I/O	-	-	A	SDMMC0_WP	I	1	PIO, I, PU, ST
								B	FLEXCOM1_IO3	I/O	1	
								D	FLEXCOM3_IO5	O	1	
								E	PWML2	O	3	
								F	FLEXCOM6_IO0	I/O	3	
AA10	AL20	VDDIOP0	GPIO	PA13	I/O	-	-	A	SDMMC0_1V8SEL	O	1	PIO, I, PU, ST
								B	FLEXCOM1_IO2	I/O	1	
								D	FLEXCOM3_IO6	O	1	
								E	PWMH2	O	3	
AA11	AL21	VDDIOP0	GPIO	PA14	I/O	-	-	F	FLEXCOM6_IO1	I/O	3	PIO, I, PU, ST
								A	SDMMC0_CD	I	1	
								B	FLEXCOM1_IO4	O	1	
								D	A25	O	1,2	
V15	AG20	VDDIOP0	GPIO	PA15	I/O	-	-	E	PWML1	O	3	PIO, I, PU, ST
								A	G0_TXEN/G0_TXCTL	O	1	
								B	FLEXCOM3_IO0	I/O	1	
								C	ISC_MCK	O	1	
								D	A1	O	1,2	
Y15	AL22	VDDIOP0	GPIO	PA16	I/O	-	-	E	TIOB0	I/O	1	PIO, I, PU, ST
								A	G0_TX0	O	1	
								B	FLEXCOM3_IO1	I/O	1	
								C	ISC_D0	I	1	
								D	A2	O	1,2	
W13	AK21	VDDIOP0	GPIO	PA17	I/O	-	-	E	TCLK0	I	1	PIO, I, PU, ST
								A	G0_TX1	O	1	
								B	FLEXCOM3_IO2	I/O	1	
								C	ISC_D1	I	1	
								D	A3	O	1,2	
Y13	AL24	VDDIOP0	GPIO	PA18	I/O	-	-	E	TIOA1	I/O	1	PIO, I, PU, ST
								A	G0_RXDV/G0_CRSDV/ G0_RXCTL	I	1	
								B	FLEXCOM3_IO3	I/O	1	
								C	ISC_D2	I	1	
								D	A4	O	1,2	
								E	TIOB1	I/O	1	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
U17	AK22	VDDIOP0	GPIO	PA19	I/O	-	-	A	G0_RX0	I	1	PIO, I, PU, ST
								B	FLEXCOM3_IO4	O	1	
								C	ISC_D3	I	1	
								D	A5	O	1,2	
								E	TCLK1	I	1	
AA12	AK24	VDDIOP0	GPIO	PA20	I/O	-	-	A	G0_RX1	I	1	PIO, I, PU, ST
								B	FLEXCOM4_IO0	I/O	1	
								C	ISC_D4	I	1	
								D	A6	O	1,2	
								E	TIOA2	I/O	1	
U15	AG21	VDDIOP0	GPIO	PA21	I/O	-	-	A	G0_RXER	I	1	PIO, I, PU, ST
								B	FLEXCOM4_IO1	I/O	1	
								C	ISC_D5	I	1	
								D	A7	O	1,2	
								E	TIOB2	I/O	1	
AA13	AL25	VDDIOP0	GPIO	PA22	I/O	-	-	A	G0_MDC	O	1	PIO, I, PU, ST
								B	FLEXCOM4_IO2	I/O	1	
								C	ISC_D6	I	1	
								D	A8	O	1,2	
								E	TCLK2	I	1	
W15	AF21	VDDIOP0	GPIO	PA23	I/O	-	-	A	G0_MDIO	I/O	1	PIO, I, PU, ST
								B	FLEXCOM4_IO3	I/O	1	
								C	ISC_D7	I	1	
								D	A9	O	1,2	
Y17	AK25	VDDIOP0	GPIO	PA24	I/O	-	-	A	G0_TXCK/G0_REFCK	I/O	1	PIO, I, PU, ST
								B	FLEXCOM4_IO4	O	1	
								C	ISC_HSYNC	I	1	
								D	A10	O	1,2	
								E	FLEXCOM0_IO5	O	1	
V17	AG22	VDDIOP0	GPIO	PA25	I/O	-	-	A	G0_125CK	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO4	O	1	
								C	ISC_VSYNC	I	1	
								D	A11	O	1,2	
								E	FLEXCOM0_IO6	O	1	
								F	FLEXCOM7_IO0	I/O	3	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
AA14	AK26	VDDIOP0	GPIO	PA26	I/O	-	-	A	G0_TX2	O	1	PIO, I, PU, ST
								B	FLEXCOM5_IO2	I/O	1	
								C	ISC_FIELD	I	1	
								D	A12	O	1,2	
								E	TF0	I/O	1	
								F	FLEXCOM7_IO1	I/O	3	
AA15	AL26	VDDIOP0	GPIO	PA27	I/O	-	-	A	G0_TX3	O	1	PIO, I, PU, ST
								B	FLEXCOM5_IO3	I/O	1	
								C	ISC_PCK	I	1	
								D	A13	O	1,2	
								E	TK0	I/O	1	
								F	FLEXCOM8_IO0	I/O	3	
T17	AF22	VDDIOP0	GPIO	PA28	I/O	-	-	A	G0_RX2	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO0	I/O	1	
								C	ISC_D8	I	1	
								D	A14	O	1,2	
								E	RD0	I	1	
								F	FLEXCOM8_IO1	I/O	3	
AA16	AL28	VDDIOP0	GPIO	PA29	I/O	-	-	A	G0_RX3	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO1	I/O	1	
								C	ISC_D9	I	1	
								D	A15	O	1,2	
								E	RF0	I/O	1	
								F	FLEXCOM9_IO0	I/O	3	
Y14	AG23	VDDIOP0	GPIO	PA30	I/O	-	-	A	G0_RXCK	I	1	PIO, I, PU, ST
								B	FLEXCOM6_IO4	O	1	
								C	ISC_D10	I	1	
								D	A16	O	1,2	
								E	RK0	I/O	1	
								F	FLEXCOM9_IO1	I/O	3	
AA17	AL29	VDDIOP0	GPIO	PA31	I/O	-	-	A	G0_TXER	O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO2	I/O	1	
								C	ISC_D11	I	1	
								D	A17	O	1,2	
								E	TD0	O	1	
								F	FLEXCOM10_IO0	I/O	3	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
T18	AF23	VDDIOP0	GPIO	PB0	I/O	-	-	A	G0_COL	I	1	PIO, I, PU, ST
								B	FLEXCOM6_IO3	I/O	1	
								C	EXT_IRQ0	I	1	
								D	A18	O	1,2	
								E	SPDIF_RX	I	2	
								F	FLEXCOM10_IO1	I/O	3	
AA18	AK28	VDDIOP0	GPIO	PB1	I/O	-	-	A	G0_CR5	I	1	PIO, I, PU, ST
								B	FLEXCOM6_IO1	I/O	1	
								C	EXT_IRQ1	I	1	
								D	A19	O	1,2	
								E	SPDIF_TX	O	2	
								F	FLEXCOM11_IO0	I/O	3	
Y19	AG24	VDDIOP0	GPIO	PB2	I/O	-	-	A	G0_TSUCOMP	O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO0	I/O	1	
								C	ADTRG	I	1	
								D	A20	O	1,2	
AA19	AL30	VDDIOP0	GPIO	PB3	I/O	-	-	F	FLEXCOM11_IO1	I/O	3	PIO, I, PU, ST
								A	RF1	I/O	1	
								B	FLEXCOM11_IO0	I/O	1	
								C	PCK2	O	2	
U19	AF24	VDDIOP0	GPIO	PB4	I/O	-	-	D	D8	I/O	1,2	PIO, I, PU, ST
								A	TF1	I/O	1	
								B	FLEXCOM11_IO1	I/O	1	
								C	PCK3	O	2	
W19	AK30	VDDIOP0	GPIO	PB5	I/O	-	-	D	D9	I/O	1,2	PIO, I, PU, ST
								A	TK1	I/O	1	
								B	FLEXCOM11_IO2	I/O	1,2,3,4,5	
								C	PCK4	O	2	
W20	AG25	VDDIOP0	GPIO	PB6	I/O	-	-	D	D10	I/O	1,2	PIO, I, PU, ST
								A	RK1	I/O	1	
								B	FLEXCOM11_IO3	I/O	1,2,3,4,5	
								C	PCK5	O	2	
								D	D11	I/O	1,2	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
W17	AK29	VDDIOP0	GPIO	PB7	I/O	-	-	A	TD1	O	1	PIO, I, PU, ST
								B	FLEXCOM11_IO4	O	1,2,3,4,5	
								C	FLEXCOM3_IO5	O	2,3,4,5	
								D	D12	I/O	1,2	
W21	AF25	VDDIOP0	GPIO	PB8	I/O	-	-	A	RD1	I	1	PIO, I, PU, ST
								B	FLEXCOM8_IO0	I/O	1	
								C	FLEXCOM3_IO6	O	2,3,4,5	
								D	D13	I/O	1,2	
Y25	AD31	VDDQSPI0	HSIO	PB9	I/O	-	-	A	QSPI0_IO3	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO1	I/O	1	
								C	PDMC0_CLK	O	1	
								D	NCS3/NANDCS	O	1	
								E	PWML0	O	2	
R20	AB28	VDDQSPI0	HSIO	PB10	I/O	-	-	A	QSPI0_IO2	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO2	I/O	1	
								C	PDMC0_DS0	I	1	
								D	NWE/NWR0/NANDWE	O	1	
								E	PWMH0	O	2	
U25	Y30	VDDQSPI0	HSIO	PB11	I/O	-	-	A	QSPI0_IO1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO3	I/O	1	
								C	PDMC0_DS1	I	1	
								D	NRD/NANDOE	O	1	
								E	PWML1	O	2	
V25	AA28	VDDQSPI0	HSIO	PB12	I/O	-	-	A	QSPI0_IO0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO4	O	1	
								C	FLEXCOM6_IO5	O	1	
								D	A21/NANDALE	O	1	
								E	PWMH1	O	2	
U22	AD30	VDDQSPI0	GPIO	PB13	I/O	-	-	A	QSPI0_CS	O	1	PIO, I, PU, ST
								B	FLEXCOM9_IO0	I/O	1	
								C	FLEXCOM6_IO6	O	1	
								D	A22/NANDCLE	O	1	
								E	PWML2	O	2	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
W25	AA30	VDDQSPI0	HSIO	PB14	I/O	-	-	A	QSPI0_SCK	O	1	PIO, I, PU, ST
								B	FLEXCOM9_IO1	I/O	1	
								D	D0	I/O	1	
								E	PWMH2	O	2	
W24	AA31	VDDQSPI0	HSIO	PB15	I/O	-	-	A	QSPI0_SCKN	O	1	PIO, I, PU, ST
								B	FLEXCOM9_IO2	I/O	1	
								D	D1	I/O	1	
								E	PWML3	O	2	
U23	W28	VDDQSPI0	HSIO	PB16	I/O	-	-	A	QSPI0_IO4	I/O	1	PIO, I, PU, ST
								B	FLEXCOM9_IO3	I/O	1	
								C	PCK0	O	1	
								D	D2	I/O	1	
								E	PWMH3	O	2	
								F	EXT_IRQ0	I	2	
T25	W31	VDDQSPI0	HSIO	PB17	I/O	-	-	A	QSPI0_IO5	I/O	1	PIO, I, PU, ST
								B	FLEXCOM9_IO4	O	1	
								C	PCK1	O	1	
								D	D3	I/O	1	
								E	PWMEXTRG0	I	2	
								F	EXT_IRQ1	I	2	
R23	W30	VDDQSPI0	HSIO	PB18	I/O	-	-	A	QSPI0_IO6	I/O	1	PIO, I, PU, ST
								B	FLEXCOM10_IO0	I/O	1	
								C	PCK2	O	1	
								D	D4	I/O	1	
								E	PWMEXTRG1	I	2	
R21	W26	VDDQSPI0	HSIO	PB19	I/O	-	-	A	QSPI0_IO7	I/O	1	PIO, I, PU, ST
								B	FLEXCOM10_IO1	I/O	1	
								C	PCK3	O	1	
								D	D5	I/O	1	
								E	PWMFI0	I	2	
U24	AB30	VDDQSPI0	HSIO	PB20	I/O	-	-	A	QSPI0_DQS	I	1	PIO, I, PU, ST
								B	FLEXCOM10_IO2	I/O	1,2,3,4,5	
								D	D6	I/O	1	
								E	PWMFI1	I	2	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
R25	V30	VDDQSPI0	GPIO	PB21	I/O	-	-	A	QSPI0_INT	I	1	PIO, I, PU, ST
								B	FLEXCOM10_IO3	I/O	1,2,3,4,5	
								C	FLEXCOM9_IO5	O	1	
								D	D7	I/O	1	
L20	P28	VDDQSPI1	GPIO	PB22	I/O	-	-	A	QSPI1_IO3	I/O	1	PIO, I, PU, ST
								B	FLEXCOM10_IO4	O	1,2,3,4,5	
								C	FLEXCOM9_IO6	O	1	
								D	NANDRDY	I	1	
K25	M31	VDDQSPI1	GPIO	PB23	I/O	-	-	A	QSPI1_IO2	I/O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO0	I/O	1	
								C	I2SMCC0_CK	I/O	1	
								F	PCK4	O	1	
L23	P30	VDDQSPI1	GPIO	PB24	I/O	-	-	A	QSPI1_IO1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO1	I/O	1	
								C	I2SMCC0_WS	I/O	1	
								F	PCK5	O	1	
L22	P25	VDDQSPI1	GPIO	PB25	I/O	-	-	A	QSPI1_IO0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO2	I/O	1	
								C	I2SMCC0_DOUT1	O	1	
								F	PCK6	O	1	
L24	N31	VDDQSPI1	GPIO	PB26	I/O	-	-	A	QSPI1_CS	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO3	I/O	1	
								C	I2SMCC0_DOUT0	O	1	
								E	PWMEXTRG0	I	1	
								F	PCK7	O	1	
L21	P26	VDDQSPI1	GPIO	PB27	I/O	-	-	A	QSPI1_SCK	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO4	O	1	
								C	I2SMCC0_MCK	O	1	
								E	PWMEXTRG1	I	1	
C17	F21	VDDSDMMC1	GPIO	PB28	I/O	-	-	A	SDMMC1_RSTN	O	1	PIO, I, PU, ST
								B	ADTRG	I	2	
								E	PWMFIO	I	1	
								F	FLEXCOM7_IO0	I/O	4	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
A19	A22	VDDSDMMC1	HSIO	PB29	I/O	-	-	A	SDMMC1_CMD	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO2	I/O	2,3,4,5	
								C	FLEXCOM0_IO5	O	2	
								D	TIOA3	I/O	1	
								E	PWMF11	I	1	
								F	FLEXCOM7_IO1	I/O	4	
A16	B21	VDDSDMMC1	HSIO	PB30	I/O	-	-	A	SDMMC1_CK	O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO3	I/O	2,3,4,5	
								C	FLEXCOM0_IO6	O	2	
								D	TIOB3	I/O	1	
								E	PWMH0	O	1	
								F	FLEXCOM8_IO0	I/O	4	
B19	B22	VDDSDMMC1	HSIO	PB31	I/O	-	-	A	SDMMC1_DAT0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO4	O	2,3,4,5	
								C	FLEXCOM9_IO5	O	2,3,4,5	
								D	TCLK3	I	1	
								E	PWML0	O	1	
								F	FLEXCOM8_IO1	I/O	4	
A18	A21	VDDSDMMC1	HSIO	PC0	I/O	-	-	A	SDMMC1_DAT1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO0	I/O	2	
								D	TIOA4	I/O	1	
								E	PWML1	O	1	
								F	FLEXCOM9_IO0	I/O	4	
								A	SDMMC1_DAT2	I/O	1	
A17	E22	VDDSDMMC1	HSIO	PC1	I/O	-	-	B	FLEXCOM3_IO1	I/O	2	PIO, I, PU, ST
								D	TIOB4	I/O	1	
								E	PWMH1	O	1	
								F	FLEXCOM9_IO1	I/O	4	
								A	SDMMC1_DAT3	I/O	1	
								B	FLEXCOM4_IO0	I/O	2	
B17	F22	VDDSDMMC1	HSIO	PC2	I/O	-	-	D	TCLK4	I	1	PIO, I, PU, ST
								E	PWML2	O	1	
								F	FLEXCOM10_IO0	I/O	4	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
D2	G1	VDDIN33	GPIO	PC3	I/O	-	-	A	SDMMC1_WP	I	1	PIO, I, PU, ST
								B	FLEXCOM4_IO1	I/O	2	
								D	TIOA5	I/O	1	
								E	PWMH2	O	1	
								F	FLEXCOM10_IO1	I/O	4	
H13	B3	VDDIN33	GPIO	PC4	I/O	-	-	A	SDMMC1_CD	I	1	PIO, I, PU, ST
								B	FLEXCOM4_IO2	I/O	2,3,4,5	
								C	FLEXCOM9_IO6	O	2,3,4,5	
								D	TIOB5	I/O	1	
								E	PWML3	O	1	
								F	FLEXCOM11_IO0	I/O	4	
C7	A3	VDDIN33	GPIO	PC5	I/O	-	-	A	SDMMC1_1V8SEL	O	1	PIO, I, PU, ST
								B	FLEXCOM4_IO3	I/O	2,3,4,5	
								C	FLEXCOM6_IO5	O	2,3,4,5	
								D	TCLK5	I	1	
								E	PWMH3	O	1	
								F	FLEXCOM11_IO1	I/O	4	
H11	L17	VDDIN33	GPIO	PC6	I/O	ACC_INP0	-	B	FLEXCOM4_IO4	O	2,3,4,5	PIO, I, PU, ST
								C	FLEXCOM6_IO6	O	2,3,4,5	
J13	F2	VDDIN33	GPIO	PC7	I/O	ACC_INN1	-	A	I2SMCC0_DIN0	I	1	PIO, I, PU, ST
								B	FLEXCOM7_IO0	I/O	2	
D1	F1	VDDIN33	GPIO	PC8	I/O	ACC_INP1	-	A	I2SMCC0_DIN1	I	1	PIO, I, PU, ST
								B	FLEXCOM7_IO1	I/O	2	
C5	D1	VDDIN33	GPIO	PC9	I/O	ACC_INN2	-	A	I2SMCC0_DOUT3	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO2	I/O	2,3,4,5	
								F	FLEXCOM1_IO0	I/O	4	
J14	G2	VDDIN33	GPIO	PC10	I/O	ACC_INP2	-	A	I2SMCC0_DOUT2	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO3	I/O	2,3,4,5	
								F	FLEXCOM1_IO1	I/O	4	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
F11	K18	VDDIN33	GPIO	PC11	I/O	ACC_INN3	-	A	I2SMCC1_CK	I/O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO4	O	2,3,4,5	
								F	FLEXCOM2_IO0	I/O	4	
B4	E1	VDDIN33	GPIO	PC12	I/O	ACC_INP3	-	A	I2SMCC1_WS	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO2	I/O	2,3,4,5	
								F	FLEXCOM2_IO1	I/O	4	
D13	K20	VDDIN33	GPIO	PC13	I/O	AD0	-	A	I2SMCC1_MCK	O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO1	I/O	2	
								F	FLEXCOM3_IO0	I/O	4	
E15	E12	VDDIN33	GPIO	PC14	I/O	AD1	-	A	I2SMCC1_DOUT0	O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO0	I/O	2	
								F	FLEXCOM3_IO1	I/O	4	
G15	F11	VDDIN33	GPIO	PC15	I/O	AD2	-	A	I2SMCC1_DOUT1	O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO3	I/O	2,3,4,5	
								F	FLEXCOM4_IO0	I/O	4	
A9	F12	VDDIN33	GPIO	PC16	I/O	AD3	-	A	I2SMCC1_DOUT2	O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO4	O	2,3,4,5	
								F	FLEXCOM4_IO1	I/O	4	
F15	E10	VDDIN33	GPIO	PC17	I/O	AD4/ RTCOUT0	-	A	I2SMCC1_DOUT3	O	1	PIO, I, PU, ST
								B	EXT_IRQ0	I	3	
								F	FLEXCOM5_IO0	I/O	4	
A8	B8	VDDIN33	GPIO	PC18	I/O	AD5/ RTCOUT1	-	A	I2SMCC1_DIN0	I	1	PIO, I, PU, ST
								B	FLEXCOM9_IO0	I/O	2	
								F	FLEXCOM5_IO1	I/O	4	
E13	F10	VDDIN33	GPIO	PC19	I/O	AD6	-	A	I2SMCC1_DIN1	I	1	PIO, I, PU, ST
								B	FLEXCOM9_IO1	I/O	2	
								F	FLEXCOM6_IO0	I/O	4	
A7	A8	VDDIN33	GPIO	PC20	I/O	AD7	-	A	I2SMCC1_DIN2	I	1	PIO, I, PU, ST
								B	FLEXCOM9_IO4	O	2,3,4,5	
								F	FLEXCOM6_IO1	I/O	4	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
C9	E11	VDDIN33	GPIO	PC21	I/O	AD8	-	A	I2SMCC1_DIN3	I	1	PIO, I, PU, ST
								B	FLEXCOM9_IO2	I/O	2,3,4,5	
								D	D3	I/O	2	
								F	FLEXCOM6_IO0	I/O	5	
A6	B7	VDDIN33	GPIO	PC22	I/O	AD9	-	A	I2SMCC0_DIN2	I	1	PIO, I, PU, ST
								B	FLEXCOM9_IO3	I/O	2,3,4,5	
								D	D4	I/O	2	
								F	FLEXCOM6_IO1	I/O	5	
A5	A7	VDDIN33	GPIO	PC23	I/O	AD10	-	A	I2SMCC0_DIN3	I	1	PIO, I, PU, ST
								B	FLEXCOM0_IO5	O	3	
								D	D5	I/O	2	
								F	FLEXCOM7_IO0	I/O	5	
C11	F9	VDDIN33	GPIO	PC24	I/O	AD11	-	B	FLEXCOM0_IO6	O	3	PIO, I, PU, ST
								C	EXT_IRQ1	I	3	
								D	D6	I/O	2	
								F	FLEXCOM7_IO1	I/O	5	
D7	B2	VDDIN33	GPIO	PC25	I/O	-	-	A	NTRST	I	1	NTRST, PU, ST
E11	E2	VDDIN33	GPIO	PC26	I/O	-	-	A	TCK_SWCLK	I	1	TCK_SWCLK, ST
B5	D2	VDDIN33	GPIO	PC27	I/O	-	-	A	TMS_SWDIO	I/O	1	TMS_SWDIO, PU, ST
H15	B4	VDDIN33	GPIO	PC28	I/O	-	-	A	TDI	I	1	TDI, PU, ST
B7	A4	VDDIN33	GPIO	PC29	I/O	-	-	A	TDO	O	1	TDO, ST
A3	B6	VDDIN33	GPIO	PC30	I/O	AD12	-	B	FLEXCOM10_IO0	I/O	2	PIO, I, PD, ST
D11	A6	VDDIN33	GPIO	PC31	I/O	AD13	-	B	FLEXCOM10_IO1	I/O	2	PIO, I, PD, ST
B1	E7	VDDIN33	GPIO	PD0	I/O	AD14	-	B	FLEXCOM11_IO0	I/O	2	PIO, I, PD, ST
F13	E8	VDDIN33	GPIO	PD1	I/O	AD15	-	B	FLEXCOM11_IO1	I/O	2	PIO, I, PD, ST
E19	B24	VDDSDMMC2	GPIO	PD2	I/O	-	-	A	SDMMC2_RSTN	O	1	PIO, I, PU, ST
								B	PCK0	O	2	
								C	CANTX4	O	1	
								D	D7	I/O	2	
								E	TIOA0	I/O	2	
								F	FLEXCOM8_IO0	I/O	5	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
A20	A24	VDDSDMMC2	HSIO	PD3	I/O	-	-	A	SDMMC2_CMD	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO0	I/O	2	
								C	CANRX4	I	1	
								D	NANDRDY	I	2	
								E	TIOB0	I/O	2	
								F	FLEXCOM8_IO1	I/O	5	
D19	F24	VDDSDMMC2	HSIO	PD4	I/O	-	-	A	SDMMC2_CK	O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO1	I/O	2	
								C	CANTX5	O	1	
								D	NCS3/NANDCS	O	2	
								E	TCLK0	I	2	
								F	FLEXCOM9_IO0	I/O	5	
C19	E25	VDDSDMMC2	HSIO	PD5	I/O	-	-	A	SDMMC2_DAT0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO2	I/O	2,3	
								C	CANRX5	I	1	
								D	NWE/NWR0/NANDWE	O	2	
								E	TIOA1	I/O	2	
								F	FLEXCOM9_IO1	I/O	5	
G17	E24	VDDSDMMC2	HSIO	PD6	I/O	-	-	A	SDMMC2_DAT1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO3	I/O	2,3	
								C	SPDIF_RX	I	3	
								D	NRD/NANDOE	O	2	
								E	TIOB1	I/O	2	
								F	FLEXCOM10_IO0	I/O	5	
F17	E23	VDDSDMMC2	HSIO	PD7	I/O	-	-	A	SDMMC2_DAT2	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO4	O	2,3	
								C	SPDIF_TX	O	3	
								D	A21/NANDALE	O	2	
								E	TCLK1	I	2	
								F	FLEXCOM10_IO1	I/O	5	
E17	F23	VDDSDMMC2	HSIO	PD8	I/O	-	-	A	SDMMC2_DAT3	I/O	1	PIO, I, PU, ST
								C	I2SMCC0_DIN0	I	2	
								D	A22/NANDCLE	O	2	
								E	TIOA2	I/O	2	
								F	FLEXCOM11_IO0	I/O	5	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
J25	N30	VDDIOP1	GPIO	PD9	I/O	-	-	A	SDMMC2_WP	I	1	PIO, I, PU, ST
								C	I2SMCC0_DIN1	I	2	
								D	D0	I/O	2	
								E	TIOB2	I/O	2	
								F	FLEXCOM11_IO1	I/O	5	
J24	N25	VDDIOP1	GPIO	PD10	I/O	-	-	A	SDMMC2_CD	I	1	PIO, I, PU, ST
								B	PCK6	O	2	
								C	I2SMCC0_DIN2	I	2	
								D	D1	I/O	2	
								E	TCLK2	I	2	
H25	L31	VDDIOP1	GPIO	PD11	I/O	-	-	F	FLEXCOM0_IO0	I/O	3	PIO, I, PU, ST
								A	SDMMC2_1V8SEL	O	1	
								B	PCK7	O	2	
								C	I2SMCC0_DIN3	I	2	
								D	D2	I/O	2	
L19	N28	VDDIOP1	GPIO	PD12	I/O	-	-	E	TIOA3	I/O	2	PIO, I, PU, ST
								F	FLEXCOM0_IO1	I/O	3	
								A	PCK1	O	2	
								B	FLEXCOM1_IO0	I/O	2	
								D	CANTX0	O	2	
K19	N26	VDDIOP1	GPIO	PD13	I/O	-	-	E	TIOB3	I/O	2	PIO, I, PU, ST
								A	I2SMCC0_CK	I/O	2	
								B	FLEXCOM1_IO1	I/O	2	
								C	PWML0	O	4	
								D	CANRX0	I	2	
J23	H31	VDDIOP1	GPIO	PD14	I/O	-	-	E	TCLK3	I	2	PIO, I, PU, ST
								A	I2SMCC0_MCK	O	2	
								B	FLEXCOM1_IO2	I/O	2,3,4	
								C	PWMH0	O	4	
								D	CANTX1	O	2	
								E	TIOA4	I/O	2	
								F	FLEXCOM2_IO0	I/O	5	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
J18	M28	VDDIOP1	GPIO	PD15	I/O	-	-	A	I2SMCC0_WS	I/O	2	PIO, I, PU, ST
								B	FLEXCOM1_IO3	I/O	2,3,4	
								C	PWML1	O	4	
								D	CANRX1	I	2	
								E	TIOB4	I/O	2	
								F	FLEXCOM2_IO1	I/O	5	
G25	K31	VDDIOP1	GPIO	PD16	I/O	-	-	A	I2SMCC0_DOUT0	O	2	PIO, I, PU, ST
								B	FLEXCOM1_IO4	O	2,3,4	
								C	PWMH1	O	4	
								D	CANTX2	O	2	
								E	TCLK4	I	2	
								F	FLEXCOM3_IO0	I/O	5	
J20	M25	VDDIOP1	GPIO	PD17	I/O	-	-	A	I2SMCC0_DOUT1	O	2	PIO, I, PU, ST
								B	FLEXCOM2_IO0	I/O	2	
								C	PWML2	O	4	
								D	CANRX2	I	2	
								E	TIOA5	I/O	2	
								F	FLEXCOM3_IO1	I/O	5	
F25	F31	VDDIOP1	GPIO	PD18	I/O	-	-	A	I2SMCC0_DOUT2	O	2	PIO, I, PU, ST
								B	FLEXCOM2_IO1	I/O	2	
								C	PWMH2	O	4	
								D	CANTX3	O	2	
								E	TIOB5	I/O	2	
								F	FLEXCOM4_IO0	I/O	5	
J21	M30	VDDIOP1	GPIO	PD19	I/O	-	-	A	I2SMCC0_DOUT3	O	2	PIO, I, PU, ST
								B	FLEXCOM2_IO2	I/O	2,3,4,5	
								C	PWML3	O	4	
								D	CANRX3	I	2	
								E	TCLK5	I	2	
								F	FLEXCOM4_IO1	I/O	5	
E25	G31	VDDIOP1	GPIO	PD20	I/O	-	-	A	PCK0	O	3	PIO, I, PU, ST
								B	FLEXCOM2_IO3	I/O	2,3,4,5	
								C	PWMH3	O	4	
								D	CANTX4	O	2	
								F	FLEXCOM5_IO0	I/O	5	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
J22	L28	VDDIOP1	GPIO	PD21	I/O	-	-	A	PCK1	O	3	PIO, I, PU, ST
								B	FLEXCOM2_IO4	O	2,3,4,5	
								D	CANRX4	I	2	
								F	FLEXCOM5_IO1	I/O	5	
								G	G1_TXEN	O	1	
G24	D31	VDDIOP1	GPIO	PD22	I/O	-	-	A	PDMC0_CLK	O	2	PIO, I, PU, ST
								C	PWMEXTRG0	I	4	
								D	RD1	I	2	
								F	CANTX5	O	2	
								G	G1_TX0	O	1	
G22	L30	VDDIOP1	GPIO	PD23	I/O	-	-	A	PDMC0_DS0	I	2	PIO, I, PU, ST
								C	PWMEXTRG1	I	4	
								D	RF1	I/O	2	
								E	ISC_MCK	O	2	
								F	CANRX5	I	2	
C25	C31	VDDIOP1	GPIO	PD24	I/O	-	-	G	G1_TX1	O	1	PIO, I, PU, ST
								A	PDMC0_DS1	I	2	
								C	PWMFI0	I	4	
								D	RK1	I/O	2	
								E	ISC_D0	I	2	
G23	K30	VDDIOP1	GPIO	PD25	I/O	-	-	G	G1_RXDV/G1_CRSDV	I	1	PIO, I, PU, ST
								A	PDMC1_CLK	O	2	
								B	FLEXCOM5_IO0	I/O	2	
								C	PWMFI1	I	4	
								D	TD1	O	2	
E23	G30	VDDIOP1	GPIO	PD26	I/O	-	-	E	ISC_D1	I	2	PIO, I, PU, ST
								G	G1_RX0	I	1	
								A	PDMC1_DS0	I	2	
								B	FLEXCOM5_IO1	I/O	2	
								C	ADTRG	I	3	
								D	TF1	I/O	2	PIO, I, PU, ST
								E	ISC_D2	I	2	
								G	G1_RX1	I	1	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
B25	E31	VDDIOP1	GPIO	PD27	I/O	-	-	A	PDMC1_DS1	I	2	PIO, I, PU, ST
								B	FLEXCOM5_IO2	I/O	2,3,4,5	
								C	TIOA0	I/O	3	
								D	TK1	I/O	2	
								E	ISC_D3	I	2	
								G	G1_RXER	I	1	
C21	F30	VDDIOP1	GPIO	PD28	I/O	-	-	A	RD0	I	2	PIO, I, PU, ST
								B	FLEXCOM5_IO3	I/O	2,3,4,5	
								C	TIOB0	I/O	3	
								D	I2SMCC1_CK	I/O	2	
								E	ISC_D4	I	2	
								F	PWML3	O	5	
C24	B30	VDDIOP1	GPIO	PD29	I/O	-	-	G	G1_MDC	O	1	PIO, I, PU, ST
								A	RF0	I/O	2	
								B	FLEXCOM5_IO4	O	2,3,4,5	
								C	TCLK0	I	3	
								D	I2SMCC1_WS	I/O	2	
								E	ISC_D5	I	2	
E22	E30	VDDIOP1	GPIO	PD30	I/O	-	-	F	PWMH3	O	5	PIO, I, PU, ST
								G	G1_MDIO	I/O	1	
								A	RK0	I/O	2	
								B	FLEXCOM6_IO0	I/O	2	
								C	TIOA1	I/O	3	
								D	I2SMCC1_MCK	O	2	
E24	A29	VDDIOP1	GPIO	PD31	I/O	-	-	E	ISC_D6	I	2	PIO, I, PU, ST
								F	PWMEXTRG0	I	5	
								G	G1_TXCK/G1_REFCK	I/O	1	
								A	TD0	O	2	
								B	FLEXCOM6_IO1	I/O	2	
								C	TIOB1	I/O	3	
								D	I2SMCC1_DOUT0	O	2	PIO, I, PU, ST
								E	ISC_D7	I	2	
								F	PWMEXTRG1	I	5	
								G	G1_TX2	O	1	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
D25	A28	VDDIOP1	GPIO	PE0	I/O	-	-	A	TF0	I/O	2	PIO, I, PU, ST
								B	FLEXCOM6_IO2	I/O	2,3,4,5	
								C	TCLK1	I	3	
								D	I2SMCC1_DOUT1	O	2	
								E	ISC_HSYNC	I	2	
								F	PWMFIO	I	5	
								G	G1_TX3	O	1	
G21	B25	VDDIOP1	GPIO	PE1	I/O	-	-	A	TK0	I/O	2	PIO, I, PU, ST
								B	FLEXCOM6_IO3	I/O	2,3,4,5	
								C	TIOA2	I/O	3	
								D	I2SMCC1_DOUT2	O	2	
								E	ISC_VSYNC	I	2	
								F	PWMF11	I	5	
								G	G1_RX2	I	1	
A23	A27	VDDIOP1	GPIO	PE2	I/O	-	-	A	PWML0	O	5	PIO, I, PU, ST
								B	FLEXCOM6_IO4	O	2,3,4,5	
								C	TIOB2	I/O	3	
								D	I2SMCC1_DOUT3	O	2	
								E	ISC_FIELD	I	2	
								G	G1_RX3	I	1	
B22	A25	VDDIOP1	GPIO	PE3	I/O	-	-	A	PWMH0	O	5	PIO, I, PU, ST
								B	FLEXCOM0_IO0	I/O	4	
								C	TCLK2	I	3	
								D	I2SMCC1_DIN0	I	2	
								E	ISC_PCK	I	2	
								G	G1_RXCK	I	1	
B21	B27	VDDIOP1	GPIO	PE4	I/O	-	-	A	PWML1	O	5	PIO, I, PU, ST
								B	FLEXCOM0_IO1	I/O	4	
								C	TIOA3	I/O	3	
								D	I2SMCC1_DIN1	I	2	
								E	ISC_D8	I	2	
								G	G1_TXER	O	1	

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
A22	A26	VDDIOP1	GPIO	PE5	I/O	-	-	A	PWMH1	O	5	PIO, I, PU, ST
								B	FLEXCOM0_IO2	I/O	4	
								C	TIOB3	I/O	3	
								D	I2SMCC1_DIN2	I	2	
								E	ISC_D9	I	2	
								G	G1_COL	I	1	
A21	B26	VDDIOP1	GPIO	PE6	I/O	-	-	A	PWML2	O	5	PIO, I, PU, ST
								B	FLEXCOM0_IO3	I/O	4	
								C	TCLK3	I	3	
								D	I2SMCC1_DIN3	I	2	
								E	ISC_D10	I	2	
								G	G1_CRS	I	1	
G20	B28	VDDIOP1	GPIO	PE7	I/O	-	-	A	PWMH2	O	5	PIO, I, PU, ST
								B	FLEXCOM0_IO4	O	4	
								C	TIOA4	I/O	3	
								E	ISC_D11	I	2	
								G	G1_TSUCOMP	I	1	
D9	K14	ADVREFP	Reference	ADVREFP	I/O	-	-	-	-	-	-	-
F14	Y22	VDDIN33	Power	VDDIN33	-	-	-	-	-	-	-	-
H16	-	GNDIN33	Ground	GNDIN33	-	-	-	-	-	-	-	-
E16	-	GNDANA	Ground	GNDANA	-	-	-	-	-	-	-	-
H10	-	GNDANA	Ground	GNDANA	-	-	-	-	-	-	-	-
H14	-	GNDANA	Ground	GNDANA	-	-	-	-	-	-	-	-
Y16	AA21	VDDIOP0	Power	VDDIOP0	-	-	-	-	-	-	-	-
Y18	AF19	VDDIOP0	Power	VDDIOP0	-	-	-	-	-	-	-	-
F24	J30	VDDIOP1	Power	VDDIOP1	-	-	-	-	-	-	-	-
H24	R28	VDDIOP1	Power	VDDIOP1	-	-	-	-	-	-	-	-
N12	A1	GND	Ground	GND	-	-	-	-	-	-	-	-
N14	A2	GND	Ground	GND	-	-	-	-	-	-	-	-
N18	A9	GND	Ground	GND	-	-	-	-	-	-	-	-
P2	A11	GND	Ground	GND	-	-	-	-	-	-	-	-
P5	A13	GND	Ground	GND	-	-	-	-	-	-	-	-
P8	A16	GND	Ground	GND	-	-	-	-	-	-	-	-
P11	A19	GND	Ground	GND	-	-	-	-	-	-	-	-
P13	A20	GND	Ground	GND	-	-	-	-	-	-	-	-
P18	A23	GND	Ground	GND	-	-	-	-	-	-	-	-
P24	A31	GND	Ground	GND	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
R1	B1	GND	Ground	GND	-	-	-	-	-	-	-	-
R3	B11	GND	Ground	GND	-	-	-	-	-	-	-	-
F10	K19	VDDANA	Power	VDDANA	-	-	-	-	-	-	-	-
F12	N15	VDDANA	Power	VDDANA	-	-	-	-	-	-	-	-
J15	N17	VDDANA	Power	VDDANA	-	-	-	-	-	-	-	-
G11	B16	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
H12	K28	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
J17	L15	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
K17	L21	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
K20	R15	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
L13	T30	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
M12	U20	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
N17	U22	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
N19	U23	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
P21	W15	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
T14	Y23	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
T16	AB20	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
T24	AF26	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
V19	AK16	VDDCORE	Power	VDDCORE	-	-	-	-	-	-	-	-
F21	F26	VDDCPU	Power	VDDCPU	-	-	-	-	-	-	-	-
G19	M23	VDDCPU	Power	VDDCPU	-	-	-	-	-	-	-	-
H20	N23	VDDCPU	Power	VDDCPU	-	-	-	-	-	-	-	-
J19	P23	VDDCPU	Power	VDDCPU	-	-	-	-	-	-	-	-
K18	R23	VDDCPU	Power	VDDCPU	-	-	-	-	-	-	-	-
R6	B13	GND	Ground	GND	-	-	-	-	-	-	-	-
R7	B20	GND	Ground	GND	-	-	-	-	-	-	-	-
R10	D30	GND	Ground	GND	-	-	-	-	-	-	-	-
R11	E6	GND	Ground	GND	-	-	-	-	-	-	-	-
R19	E13	GND	Ground	GND	-	-	-	-	-	-	-	-
T2	E21	GND	Ground	GND	-	-	-	-	-	-	-	-
T5	E26	GND	Ground	GND	-	-	-	-	-	-	-	-
T9	F6	GND	Ground	GND	-	-	-	-	-	-	-	-
T11	F20	GND	Ground	GND	-	-	-	-	-	-	-	-
T12	H1	GND	Ground	GND	-	-	-	-	-	-	-	-
T21	H2	GND	Ground	GND	-	-	-	-	-	-	-	-
U1	J31	GND	Ground	GND	-	-	-	-	-	-	-	-
U3	K1	GND	Ground	GND	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
U8	K2	GND	Ground	GND	-	-	-	-	-	-	-	-
U10	K4	GND	Ground	GND	-	-	-	-	-	-	-	-
U12	K12	GND	Ground	GND	-	-	-	-	-	-	-	-
U14	K13	GND	Ground	GND	-	-	-	-	-	-	-	-
U16	L1	GND	Ground	GND	-	-	-	-	-	-	-	-
E8	-	GNDUTMI	Ground	GNDUTMI	-	-	-	-	-	-	-	-
E10	-	GNDUTMI	Ground	GNDUTMI	-	-	-	-	-	-	-	-
E12	-	GNDUTMI	Ground	GNDUTMI	-	-	-	-	-	-	-	-
E14	-	GNDUTMI	Ground	GNDUTMI	-	-	-	-	-	-	-	-
H18	-	GNDUTMI	Ground	GNDUTMI	-	-	-	-	-	-	-	-
B8	L18	VDDUTMII	Power	VDDUTMII	-	-	-	-	-	-	-	-
B10	L19	VDDUTMII	Power	VDDUTMII	-	-	-	-	-	-	-	-
B12	M18	VDDUTMII	Power	VDDUTMII	-	-	-	-	-	-	-	-
B14	N19	VDDUTMII	Power	VDDUTMII	-	-	-	-	-	-	-	-
F16	P20	VDDUTMII	Power	VDDUTMII	-	-	-	-	-	-	-	-
A11	A10	VDDUTMI	USBPHY	HHSA_DP	I/O	-	-	-	-	-	-	-
B11	B10	VDDUTMI	USBPHY	HHSA_DM	I/O	-	-	-	-	-	-	-
C13	A18	VDDUTMI	USBPHY	HHSA_CC 1	I/O	-	-	-	-	-	-	-
D15	B19	VDDUTMI	USBPHY	HHSA_CC 2	I/O	-	-	-	-	-	-	-
A10	F13	VDDUTMI	USBPHY	HHSA_RT UNE	I	-	-	-	-	-	-	-
A13	A12	VDDUTMI	USBPHY	HHSB_DP	I/O	-	-	-	-	-	-	-
B13	B12	VDDUTMI	USBPHY	HHSB_DM	I/O	-	-	-	-	-	-	-
C15	E19	VDDUTMI	USBPHY	HHSB_CC 1	I/O	-	-	-	-	-	-	-
D17	F19	VDDUTMI	USBPHY	HHSB_CC 2	I/O	-	-	-	-	-	-	-
A12	B18	VDDUTMI	USBPHY	HHSB_RT UNE	I	-	-	-	-	-	-	-
A15	A14	VDDUTMI	USBPHY	HHSC_DP	I/O	-	-	-	-	-	-	-
B15	B14	VDDUTMI	USBPHY	HHSC_DM	I/O	-	-	-	-	-	-	-
A14	E20	VDDUTMI	USBPHY	HHSC_RT UNE	I	-	-	-	-	-	-	-
L9	P18	VDDIODDR	Power	VDDIODD R	-	-	-	-	-	-	-	-
L10	P19	VDDIODDR	Power	VDDIODD R	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
L11	R17	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
L15	R19	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
M9	T16	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
M10	T18	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
M14	T19	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
M16	U15	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
N11	V18	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
N13	V19	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
N15	W19	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
P10	Y18	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
P12	AA15	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
P14	AA17	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
R13	AA18	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
U18	L7	GND	Ground	GND	-	-	-	-	-	-	-	-
V2	L11	GND	Ground	GND	-	-	-	-	-	-	-	-
V4	L13	GND	Ground	GND	-	-	-	-	-	-	-	-
V7	L14	GND	Ground	GND	-	-	-	-	-	-	-	-
V9	M2	GND	Ground	GND	-	-	-	-	-	-	-	-
V24	M7	GND	Ground	GND	-	-	-	-	-	-	-	-
W1	M9	GND	Ground	GND	-	-	-	-	-	-	-	-
W3	M10	GND	Ground	GND	-	-	-	-	-	-	-	-
W5	M12	GND	Ground	GND	-	-	-	-	-	-	-	-
W11	M14	GND	Ground	GND	-	-	-	-	-	-	-	-
W23	M20	GND	Ground	GND	-	-	-	-	-	-	-	-
Y2	N1	GND	Ground	GND	-	-	-	-	-	-	-	-
Y5	N13	GND	Ground	GND	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
Y7	N20	GND	Ground	GND	-	-	-	-	-	-	-	-
Y10	N22	GND	Ground	GND	-	-	-	-	-	-	-	-
R12	L2	VDDIODDR	DDR	DDR_VREF	I	-	-	-	-	-	-	-
J12	T14	VDDIODDR	DDR	DDR_ZQ	I	-	-	-	-	-	-	-
T20	Y25	VDDDPHY	Power	VDDDPHY	-	-	-	-	-	-	-	-
U21	AA25	VDDDPHY	Power	VDDDPHY	-	-	-	-	-	-	-	-
V20	-	GNDDPHY	Ground	GNDDPHY	-	-	-	-	-	-	-	-
V22	-	GNDDPHY	Ground	GNDDPHY	-	-	-	-	-	-	-	-
Y21	AG30	VDDDPHY	PHY	MIPI_CLK_N	I	-	-	-	-	-	-	-
AA21	AG31	VDDDPHY	PHY	MIPI_CLK_P	I	-	-	-	-	-	-	-
Y22	AF30	VDDDPHY	PHY	MIPI_DN0	I	-	-	-	-	-	-	-
AA22	AF31	VDDDPHY	PHY	MIPI_DP0	I	-	-	-	-	-	-	-
Y20	AH30	VDDDPHY	PHY	MIPI_DN1	I	-	-	-	-	-	-	-
AA20	AH31	VDDDPHY	PHY	MIPI_DP1	I	-	-	-	-	-	-	-
AA23	AA26	VDDDPHY	PHY	MIPI_REXT	I	-	-	-	-	-	-	-
M24	Y28	VDDSDMMC0	Power	VDDSDMMC0	-	-	-	-	-	-	-	-
B18	F25	VDDSDMMC1	Power	VDDSDMMC1	-	-	-	-	-	-	-	-
Y24	P2	GND	Ground	GND	-	-	-	-	-	-	-	-
M20	V28	VDDSDMMC0	Calib pad	SDMMC0_CAL	I	-	-	-	-	-	-	-
F18	L25	VDDSDMMC1	Calib pad	SDMMC1_CAL	I	-	-	-	-	-	-	-
D24	H30	VDDSDMMC2	Power	VDDSDMMC2	-	-	-	-	-	-	-	-
B20	L26	VDDSDMMC2	Calib pad	SDMMC2_CAL	I	-	-	-	-	-	-	-
N16	-	GNDBAT	Ground	GNDBAT	-	-	-	-	-	-	-	-
K16	M22	VBAT	Power	VBAT	-	-	-	-	-	-	-	-
V13	AL12	VBAT	-	PIOBU0	I	-	-	-	-	-	-	-
T15	AF20	VBAT	-	PIOBU1	I	-	-	-	-	-	-	-
V11	AL18	VBAT	-	PIOBU2	I	-	-	-	-	-	-	-
U13	AK18	VBAT	-	PIOBU3	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
C2	J1	VDDIN33	-	XIN	I	-	-	-	-	-	-	-
C1	J2	VDDIN33	-	XOUT	O	-	-	-	-	-	-	-
Y9	AK14	VBAT	-	XIN32	I	-	-	-	-	-	-	-
AA9	AL14	VBAT	-	XOUT32	O	-	-	-	-	-	-	-
U11	AL11	VBAT	-	TST	I	-	-	-	-	-	-	-
T13	AK19	VBAT	-	JTAGSEL	I	-	-	-	-	-	-	-
Y11	AL19	VBAT	-	WKUP0	I	-	-	-	-	-	-	-
R15	AK11	VBAT	-	SHDN	O	-	-	-	-	-	-	-
G13	F8	VDDIN33	-	NRST	I	-	-	-	-	-	-	-
B9	E9	VDDIN33	-	NRST_OUT	O	-	-	-	-	-	-	-
J11	F7	VDDIN33	GPIO	AUDIOCLK	O	-	-	-	-	-	-	-
P15	AK12	VBAT	-	LPM	O	-	-	-	-	-	-	-
P20	AC30	VDDQSPI0	Power	VDDQSPI0	-	-	-	-	-	-	-	-
R17	W25	VDDQSPI0	Calib pad	QSPI0_CAL	I	-	-	-	-	-	-	-
K24	U28	VDDQSPI1	Power	VDDQSPI1	-	-	-	-	-	-	-	-
Y12	AK13	GND	Ground	GND	I	-	-	-	-	-	-	-
K12	AA19	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
K14	AB19	VDDIODDR	Power	VDDIODDR	-	-	-	-	-	-	-	-
V6	L4	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-
W2	L6	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-
W6	M1	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-
W7	N2	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-
A4	N4	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-
AA3	N6	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-
AA5	N7	DDRM_VDD	Power	DDRM_VDD	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
AA7	N9	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
B6	N10	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
E1	N12	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
E3	P1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
E7	P4	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
E9	P6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
F2	P13	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
W9	R2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
F5	R13	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
F6	T1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
F8	T4	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
G1	T6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
G6	T12	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
H2	U2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
H6	U6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
J2	U10	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
J4	U12	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
J6	V1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
Y1	V7	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
K1	V9	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
K5	V14	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
K6	W2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
K7	W4	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
K9	W6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
L2	W10	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
L4	W12	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
L6	Y1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
M1	Y9	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
M7	Y12	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
Y4	AA2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
N3	AA4	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
N5	AA6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
N6	AA11	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
N8	AB1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
P1	AB13	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
P6	AC2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
R2	AD1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
R4	AE2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
R5	AF1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
R9	AF7	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
Y6	AF12	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
T1	AG2	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
T6	AG6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
T8	AG8	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
T10	AG10	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
U2	AG11	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
U4	AG13	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
U5	AH1	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
U7	AK5	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
U9	AK7	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
V1	AL4	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
Y8	AL6	DDRM_VDD	Power	DDRM_VD D	-	-	-	-	-	-	-	-
E21	P7	GND	Ground	GND	-	-	-	-	-	-	-	-
F1	P9	GND	Ground	GND	-	-	-	-	-	-	-	-
F9	P10	GND	Ground	GND	-	-	-	-	-	-	-	-
F20	P12	GND	Ground	GND	-	-	-	-	-	-	-	-
G2	P14	GND	Ground	GND	-	-	-	-	-	-	-	-
G3	P22	GND	Ground	GND	-	-	-	-	-	-	-	-
G5	R1	GND	Ground	GND	-	-	-	-	-	-	-	-
G7	R4	GND	Ground	GND	-	-	-	-	-	-	-	-
G9	R6	GND	Ground	GND	-	-	-	-	-	-	-	-
H1	R7	GND	Ground	GND	-	-	-	-	-	-	-	-
H5	R9	GND	Ground	GND	-	-	-	-	-	-	-	-
H8	R10	GND	Ground	GND	-	-	-	-	-	-	-	-
H21	R12	GND	Ground	GND	-	-	-	-	-	-	-	-
J1	R20	GND	Ground	GND	-	-	-	-	-	-	-	-
J3	R22	GND	Ground	GND	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
J5	T2	GND	Ground	GND	-	-	-	-	-	-	-	-
J7	T7	GND	Ground	GND	-	-	-	-	-	-	-	-
J9	T13	GND	Ground	GND	-	-	-	-	-	-	-	-
J10	T20	GND	Ground	GND	-	-	-	-	-	-	-	-
J16	T31	GND	Ground	GND	-	-	-	-	-	-	-	-
K2	U1	GND	Ground	GND	-	-	-	-	-	-	-	-
K8	U7	GND	Ground	GND	-	-	-	-	-	-	-	-
K10	U9	GND	Ground	GND	-	-	-	-	-	-	-	-
K11	U13	GND	Ground	GND	-	-	-	-	-	-	-	-
K13	U17	GND	Ground	GND	-	-	-	-	-	-	-	-
K15	U19	GND	Ground	GND	-	-	-	-	-	-	-	-
K21	V2	GND	Ground	GND	-	-	-	-	-	-	-	-
L1	V4	GND	Ground	GND	-	-	-	-	-	-	-	-
L3	V6	GND	Ground	GND	-	-	-	-	-	-	-	-
L5	V10	GND	Ground	GND	-	-	-	-	-	-	-	-
L7	V12	GND	Ground	GND	-	-	-	-	-	-	-	-
L8	V13	GND	Ground	GND	-	-	-	-	-	-	-	-
L12	V20	GND	Ground	GND	-	-	-	-	-	-	-	-
L14	V22	GND	Ground	GND	-	-	-	-	-	-	-	-
L16	V23	GND	Ground	GND	-	-	-	-	-	-	-	-
L17	V31	GND	Ground	GND	-	-	-	-	-	-	-	-
L18	W1	GND	Ground	GND	-	-	-	-	-	-	-	-
M2	W7	GND	Ground	GND	-	-	-	-	-	-	-	-
M5	W9	GND	Ground	GND	-	-	-	-	-	-	-	-
M6	W13	GND	Ground	GND	-	-	-	-	-	-	-	-
M8	W17	GND	Ground	GND	-	-	-	-	-	-	-	-
M11	W20	GND	Ground	GND	-	-	-	-	-	-	-	-
M13	W22	GND	Ground	GND	-	-	-	-	-	-	-	-
M15	W23	GND	Ground	GND	-	-	-	-	-	-	-	-
M17	Y2	GND	Ground	GND	-	-	-	-	-	-	-	-
M18	Y4	GND	Ground	GND	-	-	-	-	-	-	-	-
M21	Y7	GND	Ground	GND	-	-	-	-	-	-	-	-
N1	Y10	GND	Ground	GND	-	-	-	-	-	-	-	-
N2	Y14	GND	Ground	GND	-	-	-	-	-	-	-	-
N4	Y20	GND	Ground	GND	-	-	-	-	-	-	-	-
N7	Y31	GND	Ground	GND	-	-	-	-	-	-	-	-
N9	AA1	GND	Ground	GND	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
N10	AA7	GND	Ground	GND	-	-	-	-	-	-	-	-
A1	AA13	GND	Ground	GND	-	-	-	-	-	-	-	-
A25	AA14	GND	Ground	GND	-	-	-	-	-	-	-	-
AA1	AB2	GND	Ground	GND	-	-	-	-	-	-	-	-
AA4	AB4	GND	Ground	GND	-	-	-	-	-	-	-	-
AA6	AB12	GND	Ground	GND	-	-	-	-	-	-	-	-
AA8	AB18	GND	Ground	GND	-	-	-	-	-	-	-	-
AA25	AB31	GND	Ground	GND	-	-	-	-	-	-	-	-
B2	AC1	GND	Ground	GND	-	-	-	-	-	-	-	-
B16	AC31	GND	Ground	GND	-	-	-	-	-	-	-	-
B24	AD2	GND	Ground	GND	-	-	-	-	-	-	-	-
C3	AE1	GND	Ground	GND	-	-	-	-	-	-	-	-
C23	AE30	GND	Ground	GND	-	-	-	-	-	-	-	-
D4	AE31	GND	Ground	GND	-	-	-	-	-	-	-	-
D20	AF2	GND	Ground	GND	-	-	-	-	-	-	-	-
D22	AF6	GND	Ground	GND	-	-	-	-	-	-	-	-
E2	AF8	GND	Ground	GND	-	-	-	-	-	-	-	-
E5	AF9	GND	Ground	GND	-	-	-	-	-	-	-	-
E18	AF10	GND	Ground	GND	-	-	-	-	-	-	-	-
G4	M4	DDRM_ZQ	DDR	DDRM_ZQ	I	-	-	-	-	-	-	-
E4	AB14	DDR_VREF	DDR	DDR_VREF	I	-	-	-	-	-	-	-
-	AF11	-	-	GND	-	-	-	-	-	-	-	-
-	AF13	-	-	GND	-	-	-	-	-	-	-	-
-	AG1	-	-	GND	-	-	-	-	-	-	-	-
-	AG7	-	-	GND	-	-	-	-	-	-	-	-
-	AG9	-	-	GND	-	-	-	-	-	-	-	-
-	AG12	-	-	GND	-	-	-	-	-	-	-	-
-	AG19	-	-	GND	-	-	-	-	-	-	-	-
-	AG26	-	-	GND	-	-	-	-	-	-	-	-
-	AH2	-	-	GND	-	-	-	-	-	-	-	-
-	AJ1	-	-	GND	-	-	-	-	-	-	-	-
-	AK2	-	-	GND	-	-	-	-	-	-	-	-
-	AK4	-	-	GND	-	-	-	-	-	-	-	-
-	AK6	-	-	GND	-	-	-	-	-	-	-	-
-	AK8	-	-	GND	-	-	-	-	-	-	-	-
-	AK10	-	-	GND	-	-	-	-	-	-	-	-
-	AK31	-	-	GND	-	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾ (continued)

TFBGA427	TFBGA442	Power Rail	I/O Type ⁽²⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽³⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	IO Set	
-	AL1	-	-	GND	-	-	-	-	-	-	-	-
-	AL3	-	-	GND	-	-	-	-	-	-	-	-
-	AL5	-	-	GND	-	-	-	-	-	-	-	-
-	AL7	-	-	GND	-	-	-	-	-	-	-	-
-	AL8	DDRM_VDD	-	DDRM_VDD	-	-	-	-	-	-	-	-
-	AL9	-	-	GND	-	-	-	-	-	-	-	-
-	AL10	DDRM_VDD	-	DDRM_VDD	-	-	-	-	-	-	-	-
-	AL13	-	-	GND	-	-	-	-	-	-	-	-
-	AL16	-	-	GND	-	-	-	-	-	-	-	-
-	AL23	-	-	GND	-	-	-	-	-	-	-	-
-	AL31	-	-	GND	-	-	-	-	-	-	-	-

Notes:

1. DDRM_VDD and VDDIODDR must be connected to one single-power plane of the application PCB.
2. See [Electrical Characteristics](#) for further details.
3. Signal = 'PIO' if GPIO; Dir = Direction; PU = Pull-up; PD = Pull-down; HiZ = High impedance; ST = Schmitt Trigger

10. Electrical Characteristics

The Electrical Characteristics sections in the SAMA7G5 Series and DDR3L SDRAM data sheets (see [Reference Documents](#)) apply to this device. Complementary information is provided in the following sections.

The VDDQ and VDD power inputs described in the DDR3L SDRAM data sheets are connected to the SAMA7G5 Series SiP balls called “DDRM_VDD”. Therefore, the requirements placed on VDDQ and VDD power inputs in the “Absolute Maximum Ratings” and “Recommended DC Operating Conditions” sections of these data sheets apply to the DDRM_VDD power inputs.

10.1. Recommended Thermal Operating Conditions

Table 10-1. Recommended Thermal Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
T _{J_MPU}	Junction temperature range	SAMA7G54DxG(T)-I	-40	105	°C
		SAMA7G54DxG(T)-E	-40	125	°C
T _{J_DDR3L}	Junction temperature range	SAMA7G54DxG(T)-I	-40	105	°C
		SAMA7G54DxG(T)-E	-40	125	°C

Table 10-2. TFBGA427 Package Thermal Characteristics⁽¹⁾⁽²⁾⁽³⁾

Symbol	Parameter	Typ	Unit
R _{JA}	Junction-to-ambient thermal resistance	21	°C/W

Table 10-3. TFBGA442 Package Thermal Characteristics⁽¹⁾⁽²⁾⁽³⁾

Symbol	Parameter	Typ	Unit
R _{JA}	Junction-to-ambient thermal resistance	32	°C/W

Notes:

1. $R_{JA} = (T_{J_MPU} - T_A) / P_{MPU}$, where T_A is the ambient temperature and P_{MPU} is the processor power consumption. The DDR3L SDRAM junction temperature is always lower than the MPU junction temperature.
2. According to the JEDEC JESD51-2 standard, with 2s2p board and 0 m/s air flow.
3. These values are not directly applicable to the board where the device is mounted. According to JEDEC standards, these parameters do not characterize the package itself but rather the package together with the PCB (4-layer or more) and other environmental factors (still air, etc.). For example, in still-air JEDEC-defined R_{JA} measurements, almost 70% of the power generated by the chip is dissipated from the test board, not from the package surfaces.

10.2. Power Sequences

The DDR3L SDRAM power rail (DDRM_VDD) must be connected to VDDIODDR on the PCB. Refer to Recommended Power Supply Sequencing in the Electrical Characteristics section of the SAMA7G5 Series data sheet.

10.3. Device Power Consumption in Applicative Use Cases

[Table 10-5](#) provides the device power consumption in the following conditions:

- f_{CPU_CLK}: 1 GHz
- f_{MCK1}: 200 MHz
- f_{MCK2}: 533 MHz
- f_{MCK3}: 266 MHz
- f_{MCK4}: 400 MHz

- I and D caches enabled
- Use cases run on Linux®
- Ambient temperature: 25°C
- Current consumptions are measured as shown in Figure 10-1. Note that the external component current consumptions are not counted.

Table 10-4 reports active power consumption data measured on a few SAMA7G5 SiP typical process samples. These data do not include specifications for maximum power consumption.

Table 10-4. Use Case Definition

Use Case	Description
1	Audio MP3 decoding and playback on I ² S; MP3 file on USB mass storage
2	SAMA7G54 running as iPerf server
3	Run Bonnie++ on USB mass storage
4	SAMA7G54 downloads a file from GMAC0 and copies this file to USB mass storage
5	Streaming camera on Ethernet (image format: RAW8, 1080p @ 30 fps)

Figure 10-1. Current Measurement for Applicative Use Cases

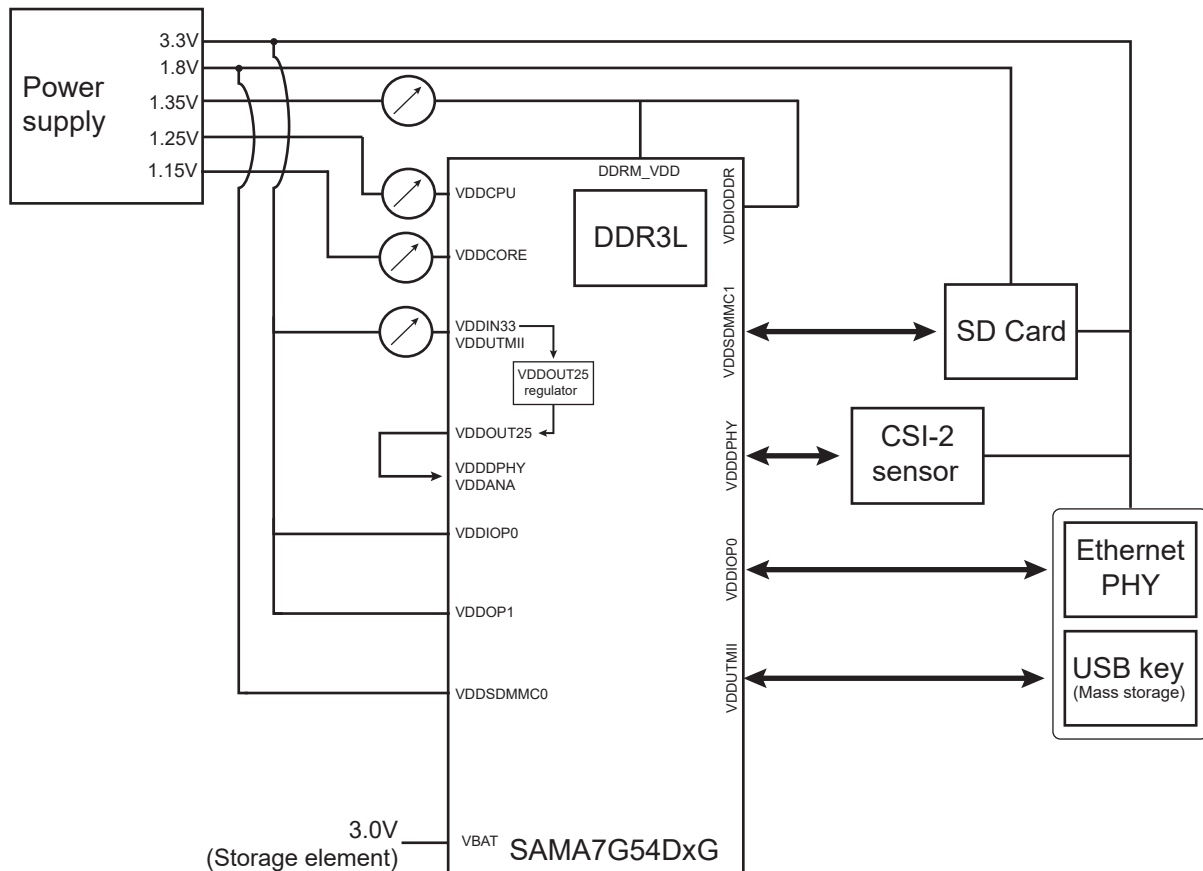


Table 10-5. Power Consumption in Applicative Use Cases

Use Case	Power Consumption (mW)				
	VDDCORE 1.15V	VDDCPU 1.25V	VDDIODDR DDRM_VDD 1.35V	VDDIN33 VDDUTMI 3.3V	Total
1	250	85	91	127	553
2	281	215	190	157	843
3	274	253	113	123	762
4	278	216	168	121	784
5	283	183	140	138	744

10.4. Power Consumption in Idle and Ultra-Low Power (ULP0, ULP1, ULP2) Modes with SDRAM in Self-Refresh

For a complete description of how to enter and exit ULP0, ULP1 or ULP2 mode, refer to the SAMA7G5 Series data sheet (see [Reference Documents](#)).

Table 10-6. Typical Power Consumption in Idle, ULP0, ULP1 or ULP2 Mode on VDDIODDR and DDRM_VDD

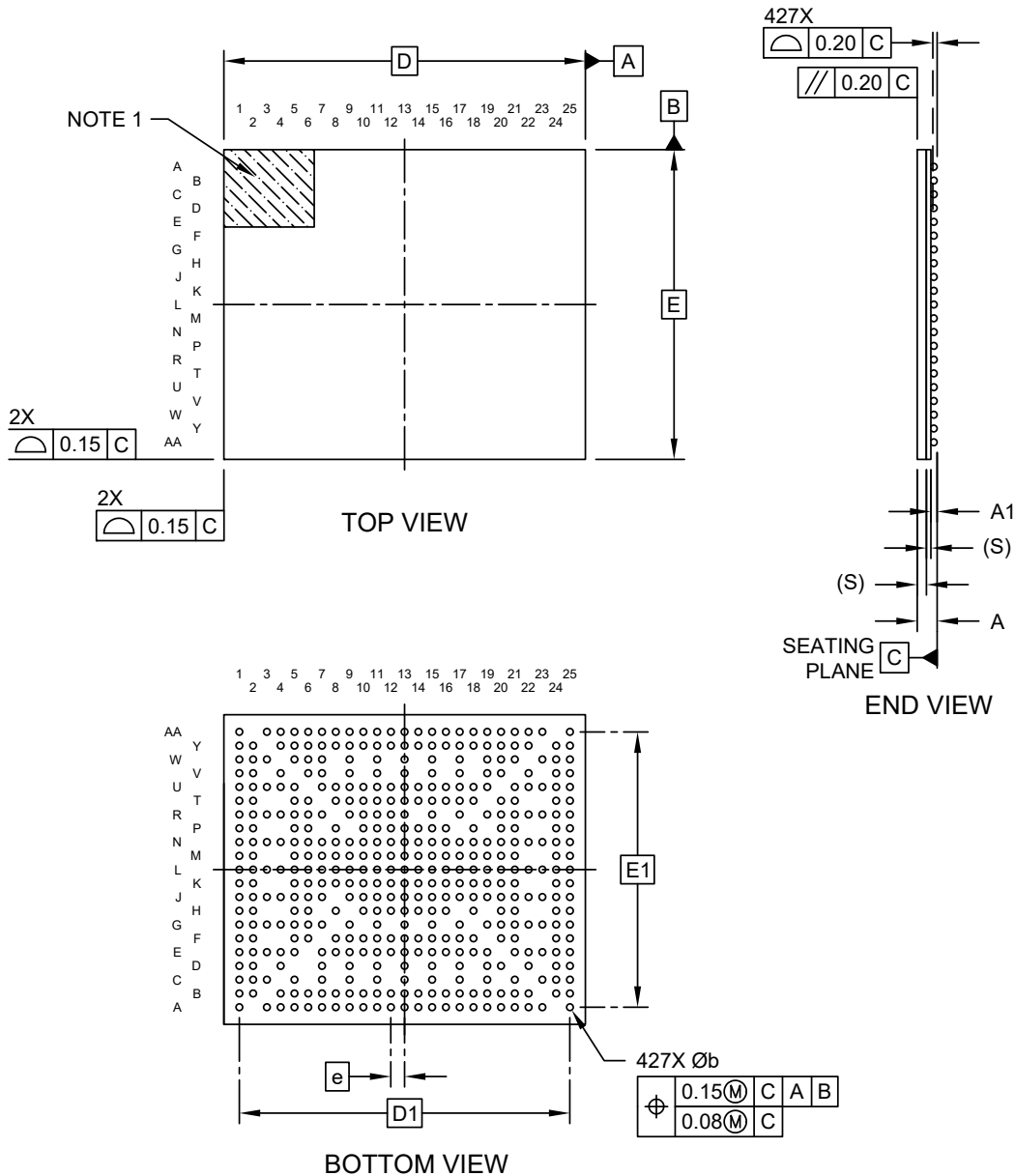
VDDIODDR DDRM_VDD (V)	T _A = -40°C	T _A = 25°C	T _A = 50°C	T _A = 70°C	T _A =85°C	Unit
1.35	3.7	4.3	5.1	6.3	7.7	mW

11. Mechanical Characteristics

11.1. 427-Ball TFBGA Mechanical Characteristics

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

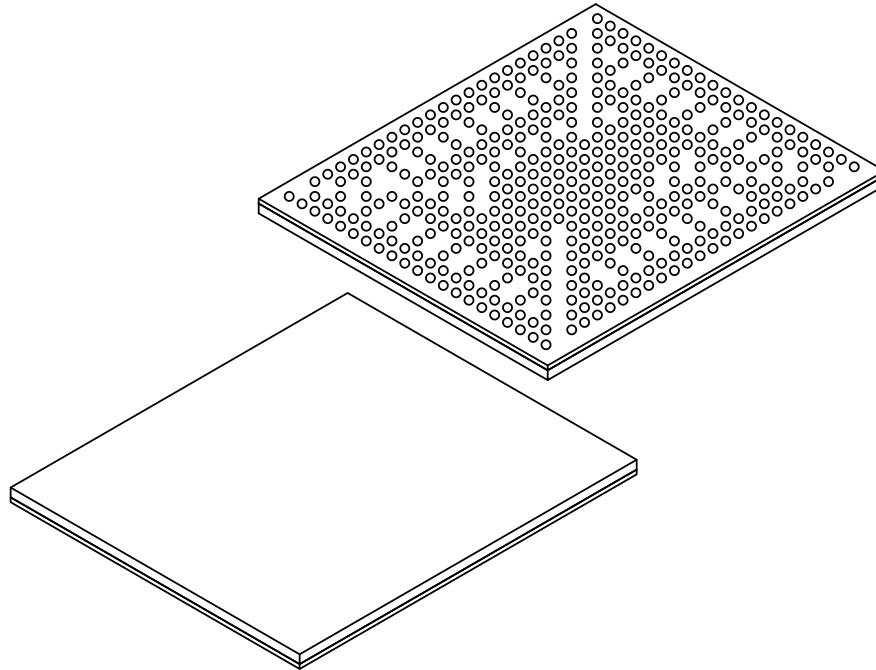
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-21537 Rev A Sheet 1 of 2

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	427		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.27	–	0.37
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.26 REF		
Overall Length	D	21.00 BSC		
Ball Array Length	D2	19.20 BSC		
Overall Width	E	18.00 BSC		
Ball Array Width	E2	16.00 BSC		
Ball Width	b	0.38	–	0.45

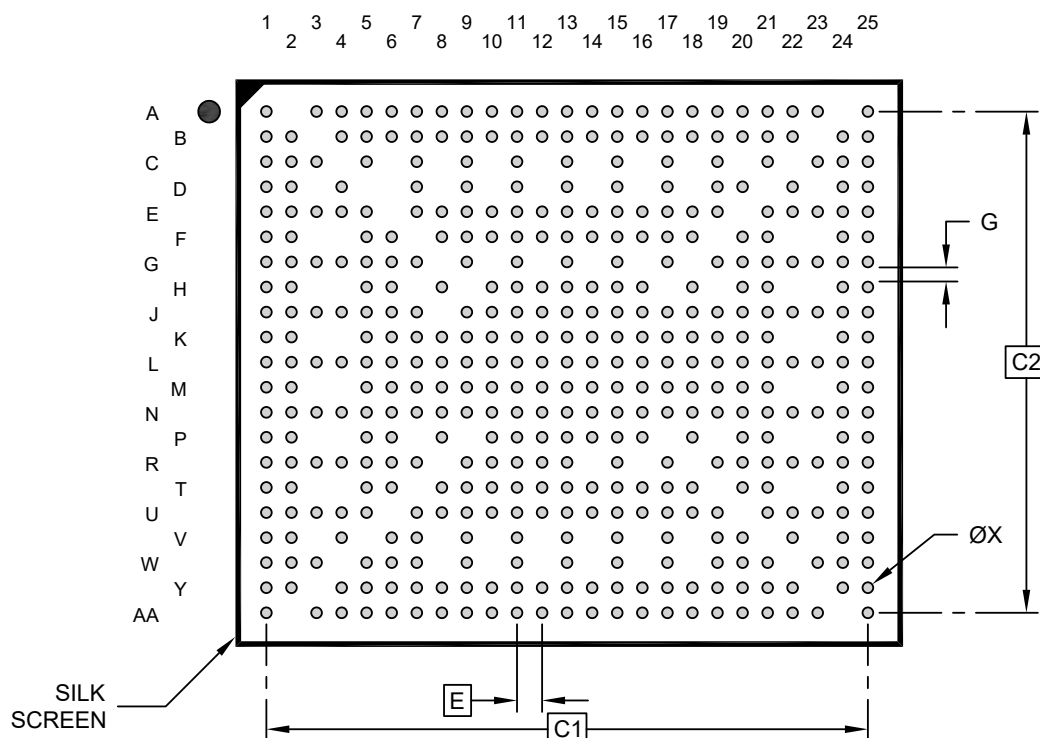
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21537 Rev A Sheet 2 of 2

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	19.20 BSC		
Contact Pad Spacing	C2	16.00 BSC		
Contact Pad Width (Xnn)	X			0.35
Contact Pad to Contact Pad (Xnn)	G	0.45		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23537 Rev A

Table 11-1. 427-ball TFBGA Package Characteristics

Moisture sensitivity level	MSL3
----------------------------	------

Table 11-2. Device and 427-ball TFBGA Package Weight

980	mg
-----	----

Table 11-3. Package Reference

JEDEC drawing reference	NA
J-STD-609 classification	e8

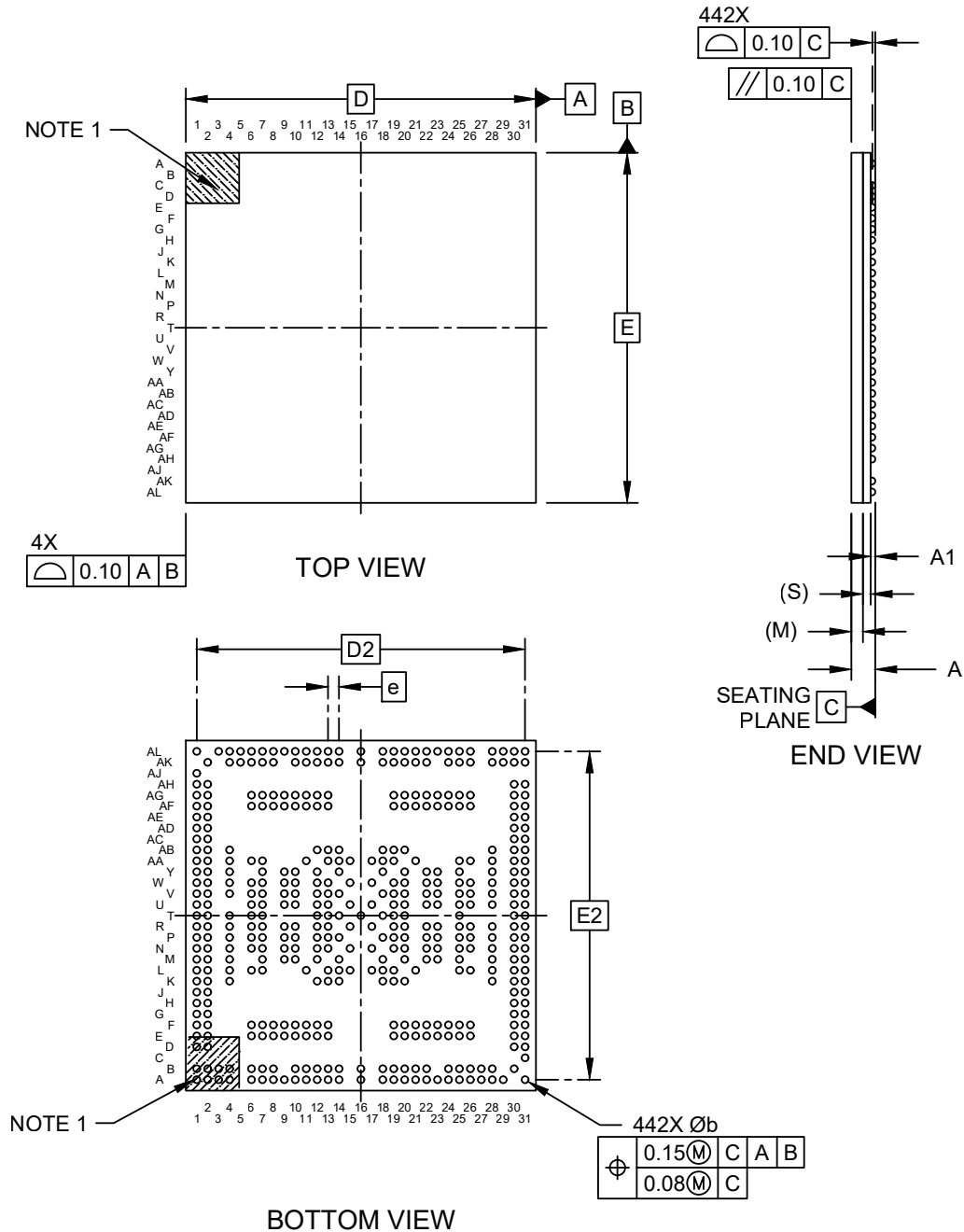
Table 11-4. 427-ball TFBGA Package Information

Ball land	0.450 mm
Nominal ball diameter	0.400 mm
Solder mask opening	0.350 mm
Solder mask definition	Solder Mask Defined (SMD)
Solder	SAC105

11.2. 442-Ball TFBGA Mechanical Characteristics

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

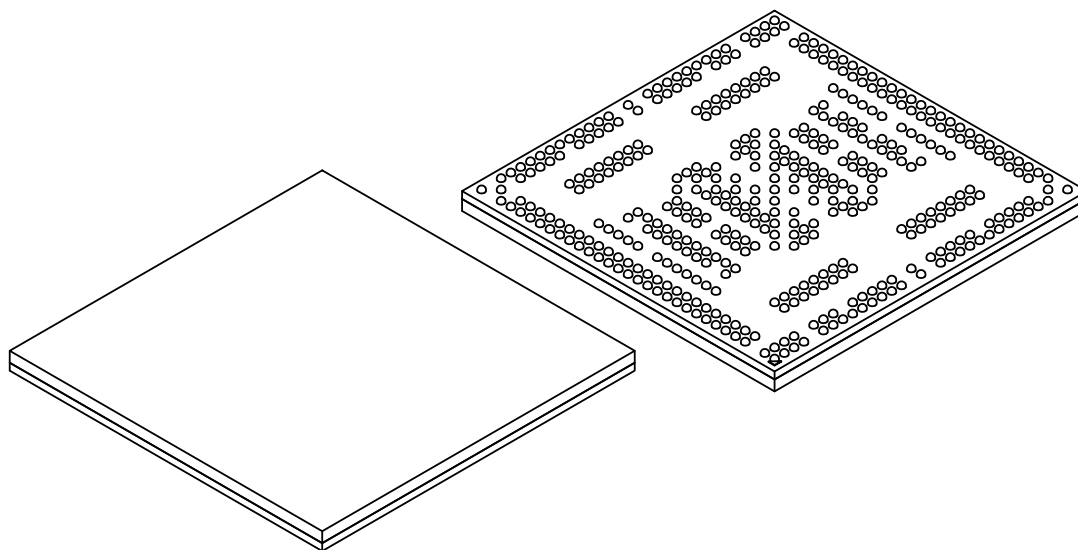
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-00686 Rev B Sheet 1 of 2

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	442		
Pitch	e	0.50 BSC		
Overall Height	A	—	—	1.20
Ball Height	A1	0.16	0.21	0.26
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.35 REF		
Overall Length	D	16.00 BSC		
Ball Array Length	D2	15.00 BSC		
Overall Width	E	16.00 BSC		
Ball Array Width	E2	15.00 BSC		
Ball Diameter	b	0.27	0.32	0.37

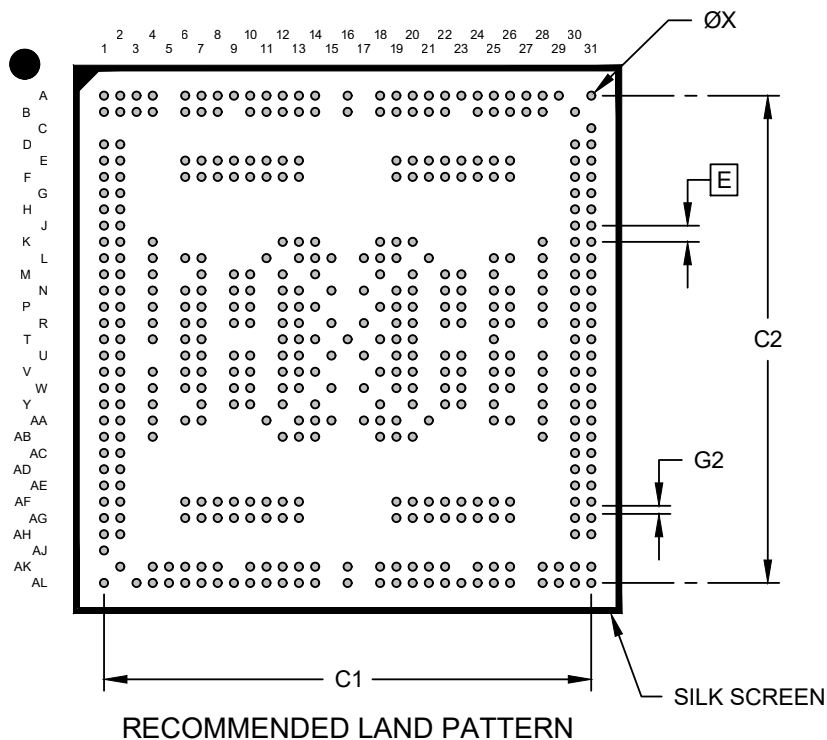
Notes:

1. The A1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
 BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
 REF: Reference Dimension, usually without tolerance, is for information purposes only.

Microchip Technology Drawing C04-00686 Rev B Sheet 2 of 2

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		15.00	
Contact Pad Spacing	C2		15.00	
Contact Pad Diameter (X442)	X			0.25
Contact Pad to Contact Pad	G2	0.25		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, please refer to the current industry standard IPC-7093.

Microchip Technology Drawing C04-02686 Rev B

Table 11-5. 442-ball TFBGA Package Characteristics

Moisture sensitivity level	MSL3
----------------------------	------

Table 11-6. Device and 442-ball TFBGA Package Weight

690	mg
-----	----

Table 11-7. Package Reference

JEDEC drawing reference	N/A
J-STD-609 classification	e8

Table 11-8. 442-ball TFBGA Package Information

Ball land	0.400 mm
Nominal ball diameter	0.320 mm
Solder mask opening	0.275 mm
Solder mask definition	Solder Mask Defined (SMD)
Solder	SAC105

12. Revision History

12.1. DS50003577E - 09/2026

Changes

Throughout: added content related to the SAMA7G54DxG(T)-I/XHB(-SLX) and SAMA7G54DxG(T)-E/XHB(-SLX)(VAO) devices

Updated [DDR3L SDRAM Features](#)

12.2. DS50003577D - 06/2025

Changes

Throughout: added content related to the SAMA7G54D4G(T)-I/4UB(-SLx) devices

12.3. DS50003577C - 12/2024

Changes

Updated [Reference Documents](#), [Ordering Information](#), [Chip Identifier](#), [Marking](#), Note 3 under [Table 10-2](#), [Table 10-6](#)

12.4. DS50003577B - 11/2023

Changes

Device renamed “SAMA7G5 Series System-in-Package (SiP)”; document title modified

Updates throughout, including in [Reference Documents](#), [Features](#), [Configuration Summary](#), [Block Diagram](#), [Chip Identifier](#)

12.5. DS50003577A - 09/2023

Changes

First issue

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Product Page Links

[SAMA7G54D1G](#), [SAMA7G54D2G](#), [SAMA7G54D4G](#)